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Research Article

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Posted Date: May 5th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9421896/v1>

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Additional Declarations: No competing interests reported.

Trafficability and excavatability of icy lunar regolith simulants quantified using cone penetration

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Abstract

This article outlines the experimental equipment and techniques used to quantify how ice content and microstructure influences the geotechnical behavior of icy lunar regolith simulants, using cone penetration tests. It addresses two primary research questions: (1) What is the relationship between ice microstructure and geotechnical behavior for icy lunar regolith simulants; and (2) How can this knowledge improve excavation and trafficability on the Moon. Unsintered, pressure sintered, ice-cemented, and vapor deposited icy regolith simulants were systematically studied at ice contents from 0.1 to 30 wt.% and bulk densities between 1.5 and 1.8 g cm⁻³. Sample and probe temperatures were maintained at -180 ± 5 °C and -160 °C to -140 °C, respectively. Trafficability was estimated using the cone index gradient (G , kPa mm⁻¹). After correcting for lunar gravity (G_{Lunar}) and experimental penetration depths to 30 mm, a vapor deposited sample with 6.68 wt.% ice was found to be 3.92 times stronger than a pressure sintered sample containing 8 wt.% ice, and 1.23 times stronger than the upper bound of *in-situ* G values measured in the Descartes Highlands during Apollo 16. The excavatability of pressure sintered samples, defined as the energy expended per unit mass of excavated water (η_{exc} , J/g), became more efficient with increasing ice content, falling from 0.35 J/g at 1 wt.% to 0.11 J/g at 30 wt.% (3.2 fold decrease), compared to an increase from 2.74 J/g at 1 wt.% to 143.03 J/g at 12 wt.% (52 fold increase) for ice-cemented samples. This work provides the first

quantitative geotechnical measurements of vapor deposited icy simulants and demonstrates that ice-cemented simulants can exhibit strength several orders of magnitude higher than other morphologies under identical conditions. If icy lunar regolith occurs as loose granular, pressure sintered or as vapor deposited morphologies these results suggest that icy lunar regolith may be easier to excavate and traverse than previously thought, given widespread use of ice-cemented simulants, rocks and concrete analogues - questioning their continued use. Consequently, excavation of water-ice within PSRs may be more economically viable, whilst also offering improved trafficability compared to dry regolith.

Keywords: Excavation, Trafficability, Icy Lunar Regolith Simulants, Cone Penetration Tests, ISRU

1 Introduction

Due to the axis of the Moon's rotation, polar topographic depressions remain in persistent shadow, called permanently shadowed regions (PSRs). Within PSRs, surface temperatures are low enough that frozen volatiles, like water-ice, can persist for billions of years due to lower sublimation rates even at ultrahigh vacuum conditions (Watson et al. 1961; Paige et al. 2010). The topic of *In-Situ* Resource Utilization (ISRU), specifically of water-ice has become a focus of academic (Mcadam et al. 2020; Kring et al. 2012), commercial (Kornuta et al. 2019), institutional (Carpenter 2019; The Space Foundation 2020) and government interest (Olszewski 2020). Yet, knowledge of the lateral and vertical distribution, quantity and composition of lunar ices also remains an area of active research (Li et al. 2018; Colaprete et al. 2010; Hayne et al. 2017). As no *in-situ* observations have been made within a PSR, the form, morphology and geotechnical behaviour of icy lunar regolith is subject to greater uncertainty. This poses significant risks for missions seeking to traverse across, excavate, or characterize deposits of dry and ice-bearing regolith in the lunar polar regions, like the Volatiles Investigating Polar Exploration Rover (VIPER) (Zacny et al. 2025; Cohan et al. 2025) or human missions as part of the Artemis program (NASA 2023; Smith et al. 2022). Highlighting this issue, the International Space Exploration Coordination Group (ISECG) identified several strategic knowledge gaps (SKGs) in their ISRU gap assessment report including; (I.D.3) Geotechnical characteristics and accessibility of PSRs, (I.D.4) Form and distribution of frozen polar volatiles, and (III.C.2) Trafficability across the lunar surface ISECG (2021).

Cone penetration tests (CPT) are a standard methodology used by industry on Earth for surface and subsurface *in-situ* geotechnical surveys of soil like clays, silts and gravel (Robertson and Cabal 2022; Lunne 1997). The cone penetrometer has a long history of predicting bearing capacity or trafficability (Shoop 1993), including off-road vehicle mobility in the NATO Reference Mobility Model (NRMM) (Ahlvin and Haley 1992; Janik and Dudziński 2024; Wong et al. 2020), studies by the U.S Army Corps (Knight 1956; Turnage 1972; Kennedy and Rush 1968), and *in-situ* measurements on the Moon during the Apollo (Mitchell et al. 1974) and Luna exploration programs (ElShafie and Chevrier 2014). Lightweight, miniaturized penetrometers with diameters

of around 3 mm to 6 mm have also been used to characterize the strength and behaviour of dry (Cil 2011; Atkinson et al. 2019; Thrift and Dreyer 2023) and icy (Mantovani et al. 2016; Atkinson et al. 2020; Johnson and Dreyer 2023) lunar regolith simulants in terrestrial laboratories (Atkinson 2023).

The mechanical performance of icy lunar regolith remains unconstrained because the physical texture and internal microstructure of ice on the Moon is not known. Several different structural polymorphs and morphologies of icy lunar regolith have been proposed (Johnson and Dreyer 2023; Gertsch et al. 2008; Šljumba et al. 2022; Sargeant et al. 2022; Rice 2000). A review of studies using icy regolith simulants found that their geotechnical behaviour can differ by several orders of magnitude, depending how they have been prepared (Ricardo et al. 2023). Sharp increases in strength were found to occur at critical ice thresholds at 1 and 3 wt.% (Atkinson et al. 2020) for ice-cemented samples made by freezing mixtures of liquid water and dry simulant. Ice enriched samples (>12 wt.%) behave like hard rock or concrete (>100 MPa) (Gertsch et al. 2008). In contrast, pressure sintered mixtures of granular ice and dry regolith at 10 wt.% are significantly weaker (<14 MPa) and did not undergo any sharp changes in strength with ice content (Johnson and Dreyer 2023). This raises questions on the suitability of different icy simulants and how this impacts validation testing of hardware intended for extraterrestrial surfaces like the Moon, Mars or comets. What has yet to be investigated in a single, controlled study is the relationship between changes in geotechnical behaviour with ice content for several morphologies of icy lunar regolith, and how this impacts trafficability and excavatability in PSRs. To address this experimental gap and the ISECG SKGs, this article outlines the experimental apparatus, procedures and results of miniature-CPT into dry, unsintered, pressure sintered, ice-cemented and vapor deposited icy regolith simulants. Ice contents from 0.1 to 30 wt.%, bulk densities between 1.5 to 1.8 g cm⁻³ and sample and probe temperatures of $-180 \pm 5^\circ\text{C}$ and -160°C to -140°C were used. The purpose of this work was to address two primary research questions: (1) What is the relationship between ice microstructure and geotechnical behavior for icy lunar regolith simulants; and (2) How can this knowledge improve excavation and trafficability on the Moon.

2 Materials and Method

2.1 Experimental Apparatus

To study the geotechnical behavior of dry and icy simulants under cryogenic conditions, a custom CPT assembly was built, shown in Figure 1. The assembly consisted of a miniature cone penetrometer attached to a low-capacity MeasureX 1 kN S-type load cell (Model MLS24), mounted to a Material Test Systems (MTS) Exceed 100 kN universal test machine (Model E45.105E). The cone was manufactured from 316 austenitic stainless steel and consisted of a 60 mm-long, 6 mm-diameter hollow probe with a 60° apex angle, mounted in a 40 mm garolite collar with a 1.5 mm T-type thermocouple inserted into the hollow cone tip. The garolite collar thermally isolated the load cell, keeping it within a few degrees of room temperature even when the probe was immersed in LN₂ (Figure 2a). The MeasureX load cell had a precision of 0.05% (± 0.5 N) and operating range of -20°C to 80°C , but no temperature compensation

was required, as variation across the assembly was measured to be $< 10^\circ\text{C}$. Calibration was done against the NATA-certified MTS Exceed 100 kN load cell. Samples were placed on a 40 mm thick garolite plate with an annular ring, secured to the MTS clevis adapter (Figure 1). This setup minimized heat conduction from the steel UTM adapter and prevented lateral movement of the sample during testing. Test parameters and boundary conditions relating cone size, container dimensions, and particle size of the regolith simulant are listed in Table 1.

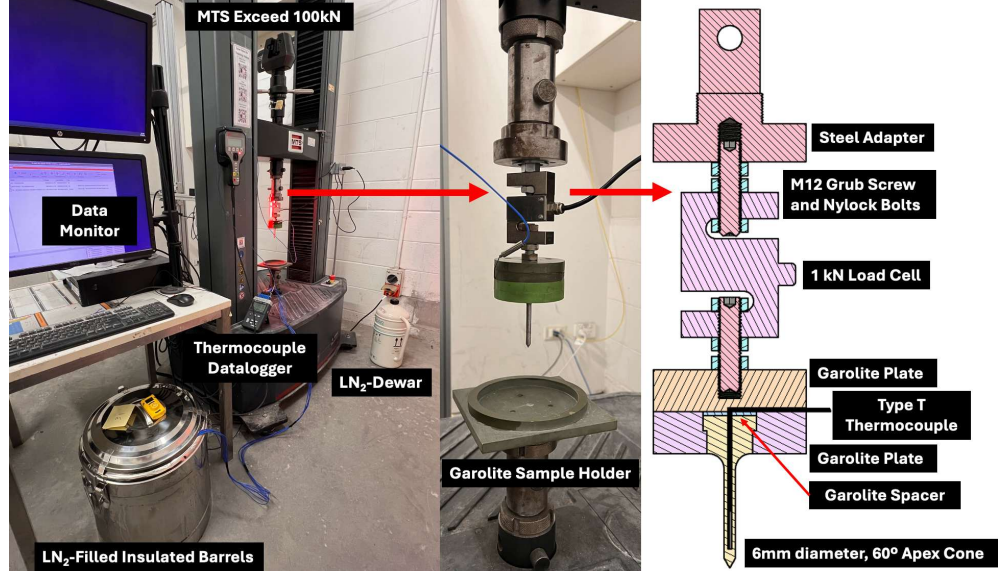


Fig. 1: Experimental setup used for cone penetration tests, including the garolite sample holder and the garolite adapter assembly for the 1 kN load cell, and a cross-sectional view of the adapter assembly.

2.2 Sample Preparation

Four types of icy lunar regolith simulant were prepared: (i) ice-cemented, (ii) unsintered, (iii) pressure sintered, and (iv) vapor deposited. Lunar highlands simulant (LHS-1) procured from [Space Resource Technologies \(2022, 2023\)](#) (formerly Exolith Lab) was used in all samples. Detailed procedures and details of the samples are given in [Ricardo \(2025\)](#). Ice-cemented samples were made by combining and slightly modifying methods from [Atkinson \(2019\)](#) and [Liu et al. \(2022\)](#), where liquid water was mixed with dry LHS-1 then compacted to a bulk density (ρ_b) of 1.5 g cm^{-3} in steel cylinders and frozen to -190°C using LN_2 . Unsintered and pressure sintered granular samples were made by modifying methods by [Johnson et al. \(2023, 2025\)](#), where shaved ice was mixed with dry LHS-1 in LN_2 before compaction. To isolate the effect of pressure induced solid-state sintering, unsintered specimens were scarified after being

Table 1: Testing parameters for cone penetrometer and boundary conditions for (I) ice-cemented, (II) unsintered and (III) pressure sintered granular ice and (IV) vapor deposited icy regolith simulant

Parameter	I to III	IV	Recommended
Container Diameter ($D_{\text{Container}}$)	104.5 mm	65 mm	–
Container Depth ($H_{\text{Container}}$)	115.8 mm	70 mm	–
Cone Diameter (D_{Cone})	6 mm	6 mm	–
Mean Particle Size (D_{50})	$87.95 \pm 6.83 \mu\text{m}^1$	$87.95 \pm 6.83 \mu\text{m}^1$	–
$D_{\text{Container}}/D_{\text{Cone}}$	17.42	10.83	10–40 ²
$H_{\text{Container}}/D_{\text{Cone}}$	19.3	11.67	>15 ³
D_{Cone}/D_{50}	68.2	68.2	>30 ⁴

¹(Long-Fox et al. 2022)

²(Kim et al. 2016; Bolton et al. 1999a; Puech and Foray 2002; Hunt et al. 2023); Kim et al. (2016) use 21.

³(Kim et al. 2016)

⁴(Bolton et al. 1999a,b)

tested, then subjected to isostatic loads of 9500^{+500}_{-700} N ($1.10^{+0.06}_{-0.08}$ MPa) for 10 min, followed by LN_2 immersion for 3 h. This is because the amount of ice and its relative distribution in the granular skeleton may greatly influence the strength (Haack et al. 2020). vapor deposited samples were prepared by controlled desublimation of water vapor into LHS-1 in a dusty thermal vacuum chamber (Ricardo et al. 2025), using smaller garolite containers (65 mm diameter, 70 mm height). Smaller sample containers were required due to limitations of the chamber volume and to achieve thermal gradients of $-6.36^\circ\text{C cm}^{-1}$ whereby ice was driven via chemical potential of vapor pressure gradients, and deposited as intergranular ice within LHS-1. Common geotechnical properties for each sample are presented in Table A1, calculated using equations from Johnson et al. (2025) using measurements of specimens repeated in triplicate. Ice content (wt_{ice}) was determined gravimetrically (ASTM 1998), whilst bulk density (ρ_b) was determined from mass and container volume. pressure sintered ρ_b was calculated as the change in sample volume after isostatic loading. Porosity (n) was calculated as shown in equation 1:

$$n = 1 - \frac{\rho_b}{(wt\%_{ice} \rho_{ice} + wt\%_{lithic} \rho_{lithic})}, \quad (1)$$

where $\rho_{ice} = 0.9168 \text{ g cm}^{-3}$ for unsintered, pressure sintered and vapor deposited ice which were measured under cryogenic conditions with crystalline ice (Feistel and Wagner 2006), and 1.0 g cm^{-3} for cemented samples measured under ambient conditions with liquid water. The specific gravity of LHS-1 was determined by Easter et al. (2022) $3.22 \pm 0.24 \text{ g cm}^{-3}$.

2.3 Experimental Procedure and Calculations

The cone penetrometer was pre-cooled in LN₂ to -190°C (Figure 2a), samples were removed from LN₂-filled barrels and loaded onto the garolite plate (Figure 2b), and the cone was positioned just above the sample surface (Figure 2c). Penetration was performed at 0.24 mm s^{-1} to a depth of 30 mm or 1000 N (whichever occurred first) with a 1 s dwell time, following the methodology of Atkinson et al. (2020). Load–displacement data were used to calculate maximum load (N), cone index (CI, kPa), and Cone Index Gradient (G , kPa mm⁻¹) using equations A1 to A4 in Appendix A.2.1. Experimental results are listed in Table A1. The elastic-viscous relaxation behavior was modelled using a two-arm Maxwell rheological model using equation A18.

CI represents the force per unit base area required to push a cone into regolith whilst G represents the relative strength of regolith across the entire penetration depth. Both have been used to estimate the trafficability regolith simulants (Daca et al. 2021; Oravec 2009; Oravec et al. 2010). To compare experimentally obtained G values (up to 30 mm depth under Earth gravity) to Apollo 15 and 16 Self Recording Penetrometer (SRP) measurements (up to 2 m depth under lunar gravity) corrections for both penetration depth and gravitational acceleration were required. Raw data for each SRP measurement was extracted from the NASA planetary data system (Mitchell et al. 2012, 2011) and cleaned following procedures described previously (Ricardo 2025, p. 229). This involved re-calculating G over the first 30 mm of penetration after normalizing depth to 0 mm, and converting G to G_{lunar} using a gravity reduction factor R_f calculated from ρ_b using equation A5 derived by Lucas et al. (2024). The predicted variation of G_{lunar} with ρ_b is given by equation A6:

$$G_{\text{lunar}} = 6.1 \times 10^{-8} e^{10.524\rho_b} \cdot (-1.65\rho_b + 3.25) \quad (2)$$

which is shown as a black-dashed line in Figure 6 which shows the change in G_{lunar} with ice content and bulk density for samples studied here. Table A3 lists the terrestrially-measured G , and corresponding R_f and G_{lunar} values for all samples. The range of Apollo 16 G values for both modified (30 mm) and unmodified penetration are shown as yellow and green shaded regions respectively, with values provided in Table A2.

To characterise the influence of ice content and microstructure on sample cohesion (c , kPa) and internal angle of friction (ϕ , $^{\circ}$), models of static cone penetration used to evaluate SRP data for Apollo 15 and 16 were used, assuming input values given in Table 2. For cohesive-frictional soils like the samples tested here, there is an infinite combination of c and ϕ values that satisfy this model, which were presented as $c - \phi$ curves in the Apollo soil mechanics reports (Mitchell et al. 1972b,a). Full details on the mathematical expressions used to obtain $c - \phi$ curves for dry and icy simulants studied here, shown in Figure 7, are derived from equations A7 to A13 using procedures outlined in Appendix A.2.3.

Once a CPT was complete, samples were manually tilted up to 110° to remove excavated material from the failure envelope on the surface of a sample. For successful inversions where the crater did not collapse (i.e. all the sample fell out), the mass of excavated material was measured using a tared scale (m_{exc} , g), with the volume of excavated material (V_{exc} , cm³) being estimated given known values of ρ_b from Table

A1. Craters were also scanned and converted to 3D meshes using photogrammetry software Polycam (2024) and processed using Autodesk Inc. (2024) to provide another estimate of excavated volume from the crater (V_{crater} , cm^3). This was intended to build on work by Gertsch et al. (2008) using new techniques. Subsequent analysis, shown in Figure A7 found that V_{crater} tended to slightly underestimate V_{exc} , and therefore only m_{exc} was used in calculations of SE and excavation efficiency. Figure 3 shows examples of excavated craters and coloured 3D meshes for pressure sintered and ice-cemented specimens.

Specific energy (SE, J cm^{-3}), the energy required to disaggregate or loosen a unit volume of material (Gertsch and Gertsch 2000), is a common metric used in terrestrial mining to quantify the efficiency of excavation (Gertsch and Gertsch 2000; Gertsch et al. 2008; Duru and Gertsch 2014). Here, SE was quantified in two ways. For loose samples that disaggregated, SE was defined as the ratio of work (W , N/m) to the volume of material displaced by the cone penetrometer (V_{disp} , cm^3), called SE of displacement (SE_{disp}). SE_{disp} is a function of cone geometry and penetration depth calculated using equation A14 and W , derived from the area under the load–displacement curve using equation A15. For samples with intact excavated craters, the SE of excavation (SE_{exc}) was calculated using the volume of excavated material (V_{exc} , cm^3) and ρ_b . To assess the energy expended to excavate a given mass of water (m_{water} , g), excavation efficiency (η_{exc} , J/g) was calculated using equations A17 and A16 given known values of $\text{wt}\%_{\text{ice}}$ and the m_{exc} using values listed in Tables A4 and A1 respectively. Thus, η_{exc} was used to evaluate the influence of ice microstructure on excavatability, and therefore, which ice deposits on the Moon would yield the highest mass of water per unit energy expended.

Table 2: Input parameters for c – ϕ curves shown in Figure 7 using model of static cone penetration for the Apollo Self Recording Penetrometers (Durgunoglu and Mitchell 1973). Values of maximum CI and ρ_b taken from (Table A1).

Parameter	This study (icy simulants)	Apollo 15 (Index 2 and 3) ¹
End bearing capacity (q_f)	Maximum Cone Index	34.5 N cm^{-2}
Effective unit soil weight (γ_s)	$\rho_b \times g_{\text{Earth}} (9.82 \text{ m s}^{-2})$	$1.8 \text{ g cm}^{-3} \times g_{\text{Moon}} (1.62 \text{ m s}^{-2})$
Cone semi-apex angle (α)	30°	15°
Soil-to-cone roughness (δ/ϕ)	0.5	0.5
Cone base diameter (B)	0.6 cm	2.024 cm
Relative depth (D/B)	(Depth corresponding to q_f)/0.6	3

¹Mitchell et al. (1972a,b); Durgunoglu and Mitchell (1973)

2.4 Experimental Design & Controls

Testing was performed in two phases. Phase 1 initially studied cemented, unsintered, and pressure sintered samples at 1, 3, 5 and 12 wt.% ice content. Ice-cemented samples with ≥ 3 wt.% ice exceeded the 1 kN load cell test capacity, whilst dry and unsintered granular samples produced forces < 10 N. A higher-capacity load cell would not resolve

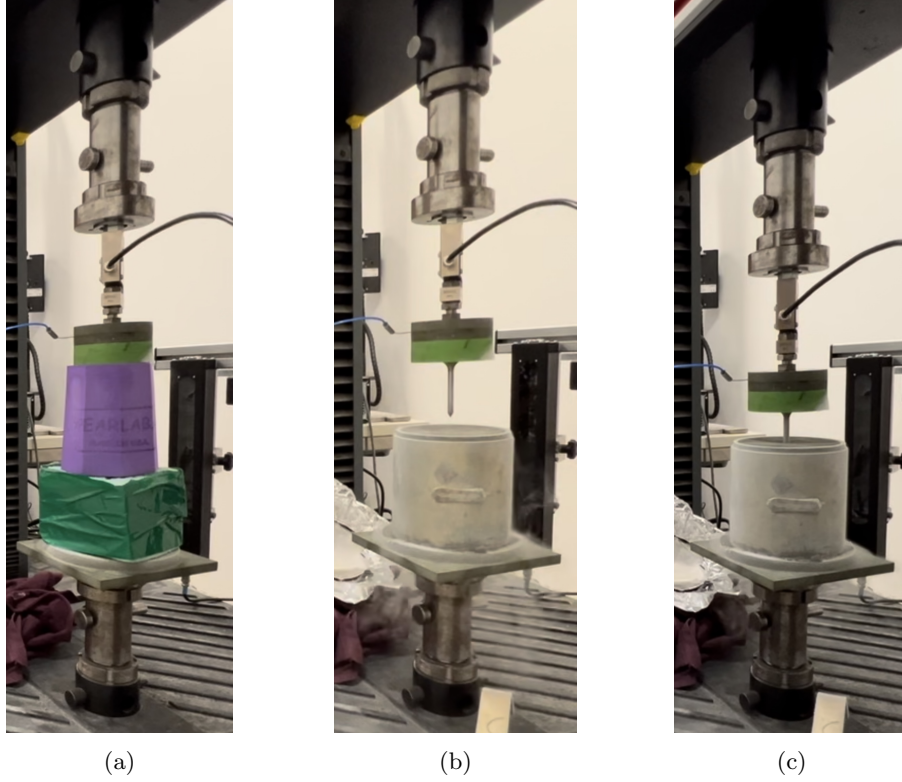


Fig. 2: Images from recording of a run showing (a) pre-cooling of the cone in LN₂, (b) a loaded sample and (c) the end of a run

weak samples, whilst a lower-capacity cell would limit testing of cemented samples. Thus, one additional 2 wt.% cemented sample and a wider range of ice contents (2 to 30 wt.%) was studied for unsintered and pressure sintered samples in Phase 2.

To minimise systematic errors, each sample was repeated in triplicate. Blank CPT runs (conducted without a sample) were also performed at the start of each experimental session, and the resulting blank load-displacement data was subtracted from raw measurements to correct for machine drift, ambient noise, and inter-experimental variability associated with changes in seasonal laboratory conditions over the year-long testing period. To address errors related to exposing samples or the probe tip to ambient laboratory conditions during testing, thermal stability of the probe tip was measured. Changes in probe temperature are shown in Figures A1b and A1a in Appendix A.1. Following removal from liquid nitrogen, the probe temperature increased rapidly to approximately -110°C within 30 s and then stabilised as it became embedded in the cold sample. During penetration and the 1 s relaxation period, cone temperatures varied by less than 5°C , stabilising near -100°C , suggesting thermal strain effects would not significantly influence measured load or displacement. In summary, it was shown that (i) the change in probe temperature

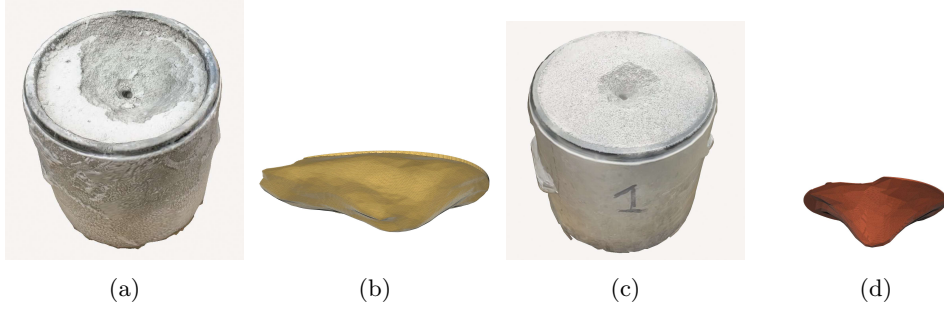


Fig. 3: Colored 3D point clouds of icy regolith samples obtained using Polycam and reconstructed meshes of the excavated crater. (a) Oblique view of the failed surface of a pressure sintered sample with 12 wt.% ice with the (b) corresponding 3D mesh with crater volume of 32.13 cm³. (c) Oblique view of the excavated crater for an ice-cemented sample containing 2 wt.% ice and (d) the corresponding 3D mesh with crater volume of 63.28 cm³. Steel cylinder has internal diameter of 104.5 mm

is highly consistent across samples in a session, and (ii) probe temperature remains stable during most of the penetration phase and throughout the relaxation phase.

3 Results

3.1 Cone Index & Cone Index Gradient

The influence of ice content and microstructure on maximum CI and G is shown in Figures 4 and 5, with values reported in Table A1. Quadratic fits provided the best representation of all datasets, with coefficients given in Table 3. The curves shown in Figures 4 and 5 are truncated, and these equations are only valid for predicting CI or G for ice content values studied here, ranging from 1 to 12, or 30 wt.% for cemented, or unsintered and pressure sintered respectively. Individual load-displacement curves for each type of icy regolith including dry LHS-1 are provided in Figures A2 through A6. The full dataset of individualized load-displacement curves for each sample are available in Online Resource 1.

There is clear evidence that ice microstructure has a significant, if not the greatest, impact on geotechnical behavior of icy lunar regolith simulants compared to changes in ice content, temperature or density. The maximum CI for dry LHS-1 compacted to 1.55 g cm⁻³ was 33.63 ± 8.69 kPa. The addition of only 1 wt.% ice increased CI by 74% (58.45 ± 14.90 kPa), 1180% (430.35 ± 80.47 kPa) and 6077% (2077.64 ± 380.33 kPa) for unsintered, pressure sintered and ice-cemented regolith respectively. At 3 wt.% ice, the range of maximum CI values differed by more than 2.5 orders of magnitude even though samples had similar density and temperatures, and were tested under identical conditions. Despite having a higher bulk density (1.76 g cm⁻³), a pressure sintered sample was more than an order of magnitude weaker than an ice-cemented sample with identical ice content (1 wt.%) and a density of 1.5 g cm⁻³. Similarly, despite having four times more ice and nearly 15% higher bulk density (1.6 g cm⁻³),

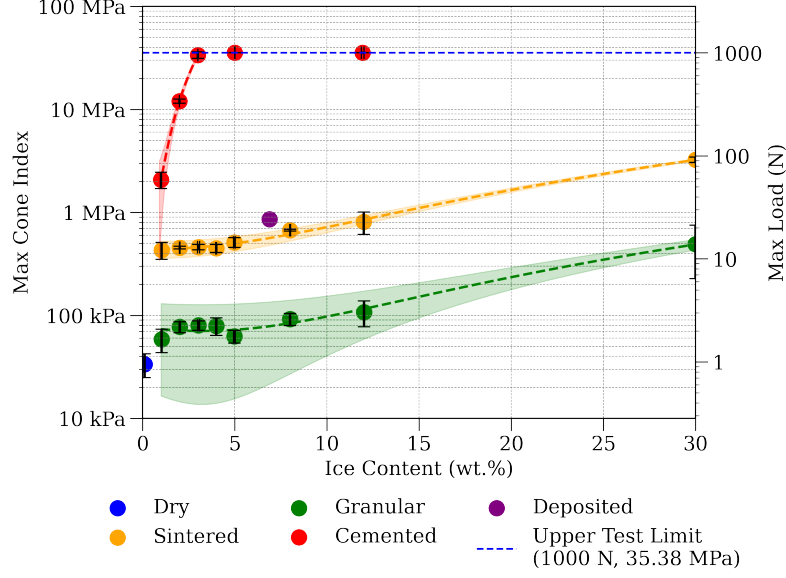


Fig. 4: Change in maximum cone index at 30 mm penetration depth with ice content (wt.%) and ice microstructure. Shaded quadratic regression represents $\pm 1\sigma$.

Table 3: Quadratic regression coefficients for change in maximum CI and G with ice content (wt.%) represented by x . Shown as dashed colored lines in Figures 4 and 5¹

Sample-Type	Maximum Cone Index (kPa)			
	$a (x^2)$	$b (x)$	$c (\text{constant})$	R^2
Sintered	3.28e+03	-5.37e+03	4.45e+05	0.9882
Granular	5.82e+02	-3.73e+03	7.65e+04	0.8066
Cemented	5.86e+06	-7.58e+06	3.82e+06	0.9901

Sample-Type	Cone Index Gradient (G , kPa/mm)			
	$a (x^2)$	$b (x)$	$c (\text{constant})$	R^2
Sintered	16.4	-0.137	0.0911	0.962
Granular	2.79	-0.16	0.0199	0.852
Cemented	-1.94e+03	1.55e+03	-53.2	0.928

¹Only valid for predicting CI or G for ice content values measured here, between 1 to 12, or 30 wt.% for cemented, or unsintered and pressure sintered samples respectively.

a pressure sintered sample containing 12 wt.% ice was 97.6% (42-times) weaker than an ice-cemented sample with only 3 wt.% ice and density of 1.5 g cm^{-3} .

Differences in the relationship between ice content and geotechnical behavior were also observed. Similar to observations by [Atkinson et al. \(2020\)](#), the addition of small (1 to 3 wt.%) amounts of ice into dry regolith caused order-of-magnitude increases in

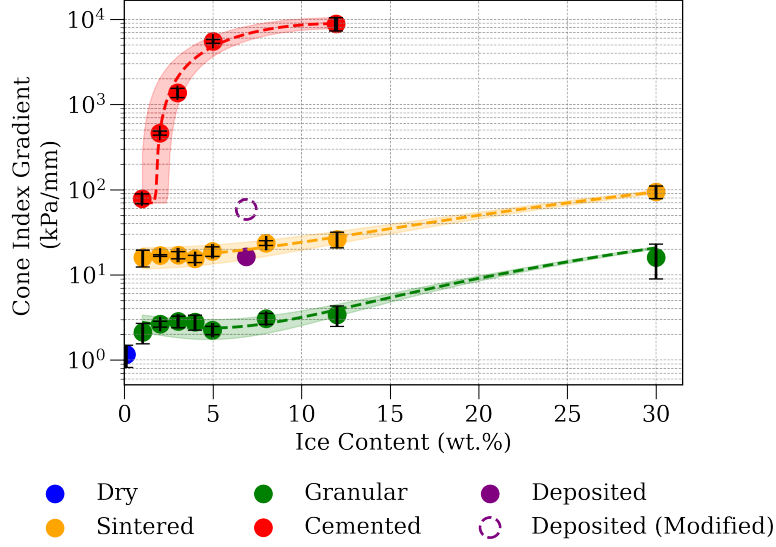


Fig. 5: Change in Cone Index Gradient, G (kPa/mm), with ice content (wt.%) and ice microstructure. Shaded quadratic regression represents $\pm 1\sigma$. The filled and dashed purple circles represent the original and modified G respectively, calculated from either the full data or the initial ramp in load to the maximum recorded cone index value for vapor deposited icy regolith (i.e G of the ice lens not the entire sample)

maximum CI for ice-cemented samples. In contrast, unsintered and pressure sintered samples required 30 wt.% ice to achieve an order-of-magnitude increase in strength. This indicates the strength of granular icy regolith being less sensitive to changes in ice content compared to ice-cemented microstructures, due to differences in the granular skeleton and physical bonding (or lack of) of ice and lithic grains. This provides quantitative evidence supporting qualitative observations of ice microstructure observed using micro-computed tomography (Micro-CT) performed in parallel to this work (Ricardo et al. 2026), discussed further in Section 4. Although pressure sintered samples had different bulk densities due to ice being measured and prepared relative to mass rather than volume, this could affect interpretation of obtained results. However, samples containing 1 to 5 wt.% ice had G_{lunar} nearly six-times higher than dry LHS-1 at comparable densities between 1.70 g cm^{-3} to 1.75 g cm^{-3} (Lucas et al. 2024), indicating that the presence and sintering of ice grains contributed to elevated strength, more than differences in bulk density alone.

The predicted trafficability of PSRs evaluated using G also varied widely, depending on ice microstructure and ice content. All ice-cemented samples, even those with 1 wt.% ice, exhibited G_{Lunar} values 63.2% (1.63-times) higher than the Apollo 16 upper limit, even after correcting to 30 mm penetration depth. In contrast, pressure sintered samples require at least 5 wt.% ice to approach the Apollo 16 lower limit, whilst a vapor deposited sample containing 6.88 wt.% ice exhibited intermediate strength, being 85% (1.85-times) stronger than the Apollo 16 lower limit. When only considering the initial

ramp to the depth of the observed ice lens (≈ 15 mm) in the vapor deposited specimen (Figure A6), the modified G_{Lunar} was much stronger, being 22.57% (1.23 times) higher than the Apollo 16 upper limit.

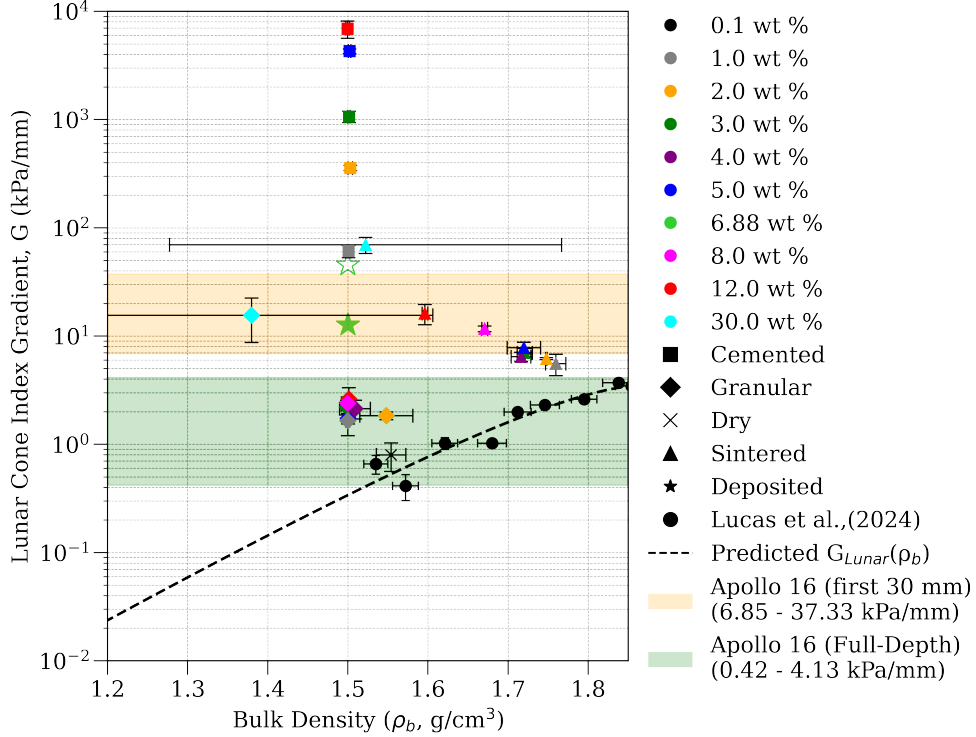


Fig. 6: Change in G slope parameter as a function of bulk density (ρ_b), ice content and ice microstructure for dry and icy lunar regolith simulants under lunar gravity (G_{Lunar}) using the reduction factor calculated from Lucas et al. (2024). G values of dry LHS-1 using CPT at various densities given by Lucas et al. (2024) and the predicted G_{Lunar} as function of ρ_b given as $G_{Lunar} = 6.110^{-8} e^{10.524\rho_b} / (-1.65\rho_b + 3.25)$. Shaded green and orange regions represent range of G values measured *in-situ* during Apollo 16 over the full depth range and the first 30 mm of depth respectively. The modified G_{Lunar} for the vapor deposited sample calculated to the depth of the ice lens (15 mm) and full depth (30 mm) shown by the dashed and fully colored green star respectively

3.2 Cohesion and Friction

The influence of ice content and microstructure on the cohesion and friction, presented as $c - \phi$ curves, are shown in Figure 7. Ice-cemented samples exhibited up to 2 orders of magnitude higher c than other morphologies of icy regolith - even with identical ice content. At 30 wt.% ice, sintered and unsintered samples were weaker (interpreted as

left translated $c - \phi$ curve) than an ice-cemented sample with only 2 wt.% ice, which
 was still stronger than measurements at the base of the Apollo 15 trench. The vapor
 deposited sample had strength characteristics approximately analgous to the base of
 the Apollo 15 trench, with intermediate strength between that of sintered and ice-
 cemented regolith. Compared to Apollo regolith, dry LHS-1 had lower c and ϕ values
 likely due to the absence of agglutinates and other ultra-fine, highly angular particles
 that contribute to its modified 'curved' Mohr-Coulomb shear behavior [Carrier et al.](#)
[\(1973, 1972\)](#); [Leonovich et al. \(1977\)](#). For comparison, the $c - \phi$ curve of the average
 of two SRP measurements on Apollo 15 taken adjacent to, and at the bottom of
 the soil mechanics trench (Index 2 and 3) is shown as a dashed blue line, as well as
 recommended values of c and ϕ for Apollo regolith at different depths ([Carrier III et al.](#)
[1991](#); [Durgunoglu and Mitchell 1973](#)). There is good agreement between the predicted
 $c - \phi$ curve of LHS-1 tested here compared to c and ϕ values for LHS-1 measured using
 direct shear ([Long-fox et al. 2023](#)) and derived by [Dotson et al. \(2023\)](#), indicating this
 approach is valid for estimating c and ϕ for icy regolith simulants.

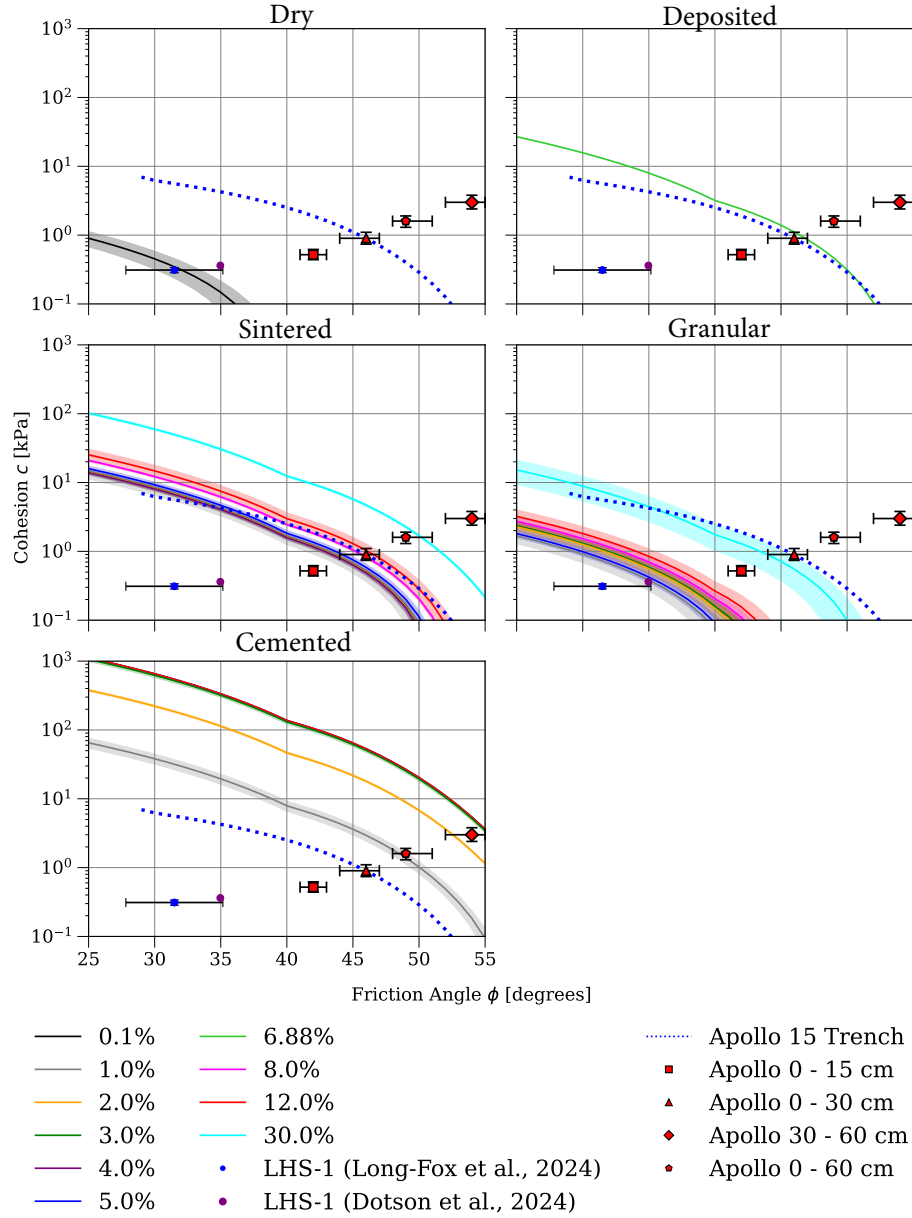


Fig. 7: Change in cohesion and friction angle with ice content (wt.%) for dry LHS-1, vapor deposited, pressure sintered and unsintered granular, and ice-cemented regolith simulants compared to Apollo data (Carrier III et al. 1991; Durgunoglu and Mitchell 1973). Curves represent average of values taken in triplicate with shaded region representing $\pm 1\sigma$.

3.3 Excavation Efficiency

The influence of ice content and microstructure on SE_{disp} , SE_{exc} , and η_{exc} is shown in Figure 8 for dry and icy lunar regolith simulants. Quadratic and hyperbolic fits best describe the variation in excavation efficiency with ice content for cemented and pressure-sintered samples, respectively. Corresponding regression coefficients are listed in Table 4. A complete divergence in η_{exc} as a function of ice content was observed, highlighting the importance of considering ice microstructure when predicting excavatability. The η_{exc} of ice-cemented samples can be characterized using the equation:

$$\eta_{exc}(wt_{ice}) = -4.13wt_{ice}^2 + 67.9wt_{ice} - 77.8 \quad (3)$$

where efficiency decreases with increasing ice content, to a peak near saturation (least efficient excavatability) which reduces as the material further increases in ice content. This ratifies similar observations of ice-cemented samples (Liu et al. 2024; Tsyrovich 1975), defined previously as a strength saturation arc (Ricardo et al. 2024) where changes in strength occur due to transitions from the pendular, funicular and capillary saturation regimes during sample preparation prior to freezing. The η_{exc} of pressure sintered samples was characterized by a hyperbolic function as:

$$\eta_{exc}(wt_{ice}) = \frac{0.313}{wt_{ice}} + 0.0356 \quad (4)$$

where efficiency increases with ice content. These results confirm that excavation efficiency depends strongly on the chosen icy simulant, leading to vastly different experimental outcomes. If ice-cemented materials are used to simulate the LCROSS impact site (5 wt.% ice), one would expect high excavation forces with lower water recovery corresponding to poor energy efficiency per unit mass of H_2O extracted compared to regolith containing <3 wt.% ice. In contrast, pressure sintered materials exhibit the opposite trend, where increasing ice content leads to improved excavation efficiency and greater water yield.

Equations 3 and 4 may be used to predict excavatability of icy lunar regolith, but with several limitations. The equations are only valid over the range of ice content studied here being 1 to 12 wt.% for ice-cemented and 1 to 30 wt.% for sintered samples. This means behaviour of low (<1 wt.%) or highly concentrated deposits (> 30 wt.%) is uncertain. The quadratic coefficient in equation 3 used to calculate η_{exc} for ice-cemented samples also becomes negative where the ice content is of 1.24 and 15.2 wt.% (i.e. where the curves intersect the x-axis), implying energy expenditure becomes negative, which is invalid. Additionally, ice-cemented samples with $wt_{ice} > 12wt\%$ exceeded the 1 kN test capacity after only < 5mm of penetration - leaving a small conical indent on the sample surface. Although the measured mass of material after inversion was about < 1 g, it is likely to be excess material and dust on the sample surface, not material that was actually excavated. For this reason, the mass of excavated material was approximated as 0.01 g, to avoid division by zero for η_{exc} calculations.

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Table 4: Quadratic and hyperbolic regression coefficients for specific energy of displacement, specific energy of excavation, and excavation efficiency with ice Content. x represents ice content (wt.%).

Specific Energy of Displacement (SE_{disp} , J/cm ³)				
Sample-Type	a (x ²)	b (x)	c (constant)	R ²
Sintered	0.00111	-0.00591	0.244	0.914
Granular	0.000227	-0.00151	0.0378	0.858
Cemented	-0.394	8.75	-7.93	0.97

Specific Energy of Excavation (SE_{exc} , J/cm ³)				
Sample-Type	a (x ²)	b (x)	c (constant)	R ²
Cemented	-0.0991	3.76	-5.1	0.941
Sintered	4.87e-05	-1.41e-05	0.00606	0.9

Excavation Efficiency (η_{exc} , J/g)				
Sample-Type	a (x ²)	b (x)	c (constant)	R ²
Cemented	-4.13	67.9	-77.8	0.89

	a (constant)	b (1/x)	R^2
Sintered	0.0356	0.313	0.78

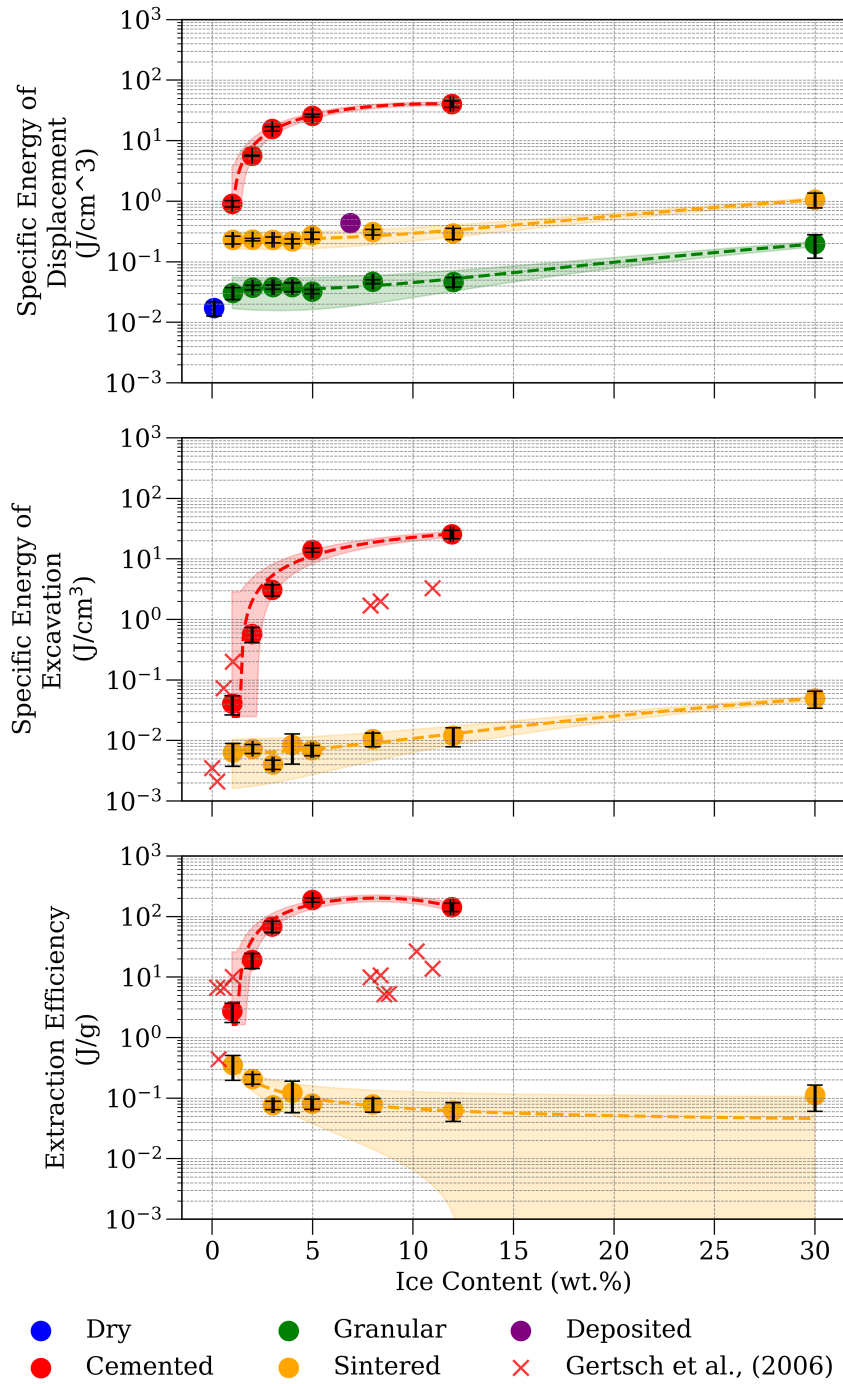


Fig. 8: (Top) Effect of ice content and microstructure on specific energy of displaced volume (cm^3), (middle) specific energy of excavation (J cm^{-3}), and (bottom) excavation efficiency (J/g). Dashed line indicate regression fit with shaded region representing $\pm 1\sigma$.

3.4 Modeling Relaxation Behavior

The variation in relaxation model parameters with ice content and microstructure for all samples is shown in Figure 9, fitted using Equation A18. Outliers were removed using a Z-score threshold of ± 10 . This threshold offered the best balance between the need to remove substantial outliers without excessive removal of valid data points, which could risk over-fitting the dataset. Ice-cemented samples exhibited the lowest elasticity (k_1 and k_2) values which also decreased with increasing ice content in agreement with observations of ice-cemented samples by Atkinson et al. (2020, Fig. 7). This is interpreted as a decreasing capacity to support residual load meaning higher stiffness (brittleness). This behaviour is due to sharply increasing cohesive strength, as the walls of pore cavities and grain contacts become covered then filled with cemented water-ice, and is supported by similar observations from $c - \phi$ curves in Figure 7. In contrast, unsintered and pressure sintered samples exhibited more elastic behaviour, with pressure sintered samples having higher elasticity and lower strain (ϵ), whilst vapor deposited samples had the highest k_1 and k_2 , and lowest τ_1 , τ_2 and ϵ , indicating slower relaxation, higher elasticity and lowest strain. Vapor deposited ice can therefore sustain smaller residual load but exhibits a more elastic, but slow relaxation response.

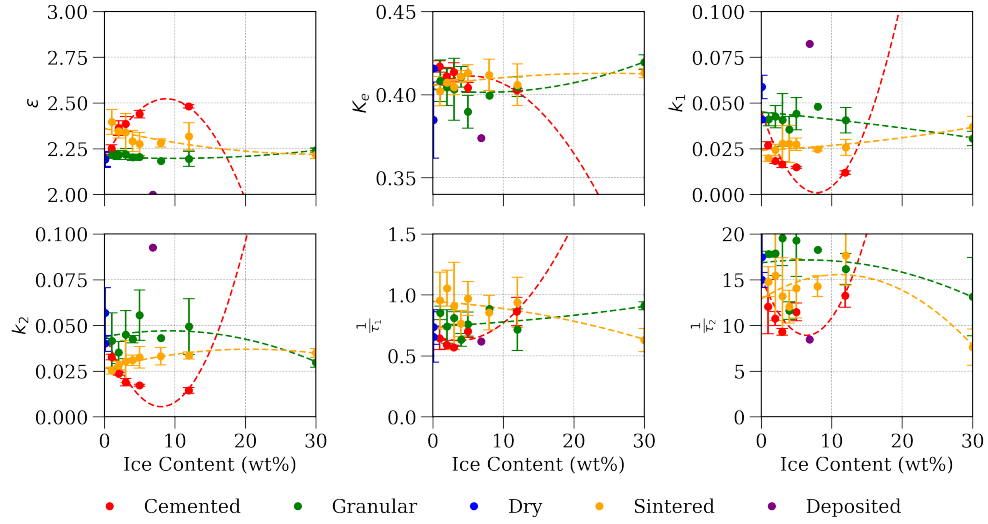


Fig. 9: Parameters of the two-arm Maxwell model fits to relaxation curve using equation A18 after 1 sec of relaxation as a function of ice content and ice microstructure.

4 Discussion

The experimental results presented here indicate that the geotechnical behaviour of icy lunar regolith simulants are primarily a function of ice microstructure, which in turn is dependant on the physical state of water during sample preparation. Although the presence of ice, regardless of its microstructure, increases regolith strength, the extent of the increase can vary by several orders of magnitude. This has direct implications for predicting the performance of trafficability and excavatability within PSRs.

4.1 Ice-cemented regolith

In all metrics (CI , G_{Lunar} , $c - \phi$, relaxation behavior), ice-cemented samples exhibited distinct geotechnical behaviour to other icy simulants, with significantly elevated strength and brittleness with increasing ice content. The transition from 1 to 3 wt% ice corresponded to order-of-magnitude increases in CI and G_{Lunar} (Figures 4 and 5). Samples containing 5 and 12 wt.% ice exceeded the test capacity of the 1 kN load cell. Similar observations were reported by [Atkinson et al. \(2020\)](#). G_{Lunar} values increased by 1649% from 1 to 3 wt.%, with samples containing 1 wt.% and 3 wt.% being 63% (1.6-times) or 855% (9.6-times) stronger than the upper limit of Apollo 16 SRP measurements respectively.

Due to its high strength, ice-cemented samples are often compared with rock or concrete. These comparisons are made using unconfined compressive strength (UCS), Brazilian tensile strength (BTS), or flexural strength in a 3-point bend test, which are common metrics used in terrestrial mining and construction ([Zacny et al. 2007](#); [Gertsch and Gertsch 2000](#); [Gertsch et al. 2008](#); [Liu et al. 2022, 2024](#); [Ma et al. 2023](#); [Ash et al. 2016](#)). Because of this, rock or concrete have been used to simulate ice-cemented samples - essentially acting a simulant of a simulant ([Joshi 2021](#); [Stoker et al. 2024](#); [Atkinson 2016](#); [Gertsch et al. 2008](#)). This may reinforce assumptions that icy lunar regolith is extremely strong, like rock, and will require percussive ([Zacny et al. 2022](#)) or ballistic sampling techniques ([Ma et al. 2023](#)). Notably, participants in NASA's Break the ice Challenge used large discrete- or continuous-bucket excavators and jackhammers to excavate low (1.5 to 2 MPa) and moderate (20 to 35 MPa) strength concrete that was used to simulate icy regolith containing 4 wt.% and 10 wt.% ice respectively ([Leucht et al. 2025](#); [Rusch 2023](#); [Vetcha et al. 2022](#); [Johnson et al. 2023](#)). It should be acknowledged that instruments designed for lunar excavation and prospecting must accommodate a range of materials, including buried rocks ([Feng et al. 2022](#)) and dense ($>2.0 \text{ g cm}^{-3}$) regolith at depth ([Carrier III et al. 1991](#)). However, significant size, mass and power savings may be possible if lower power motors or excavation techniques ([Just et al. 2020](#)) are deemed sufficient to excavate weaker granular or vapor deposited icy regolith, as results here suggest.

In previous work ([Ricardo et al. 2023](#)), we proposed that the large differences in strength for ice-cemented samples are due to changes in the granular skeleton during sample preparation at different water content, which correspond to different soil saturation regimes ([Capes 1980](#); [Newitt and Conway-Jones 1958](#)). Namely, the transition through pendular (dry, $<1 \text{ wt.}\%$), funicular (slightly wet, 1 to 3 wt.%) and capillary (wet, $>5 \text{ wt.}\%$) regimes. Visualising this, Figure 10 shows Micro-CT images of LHS-1

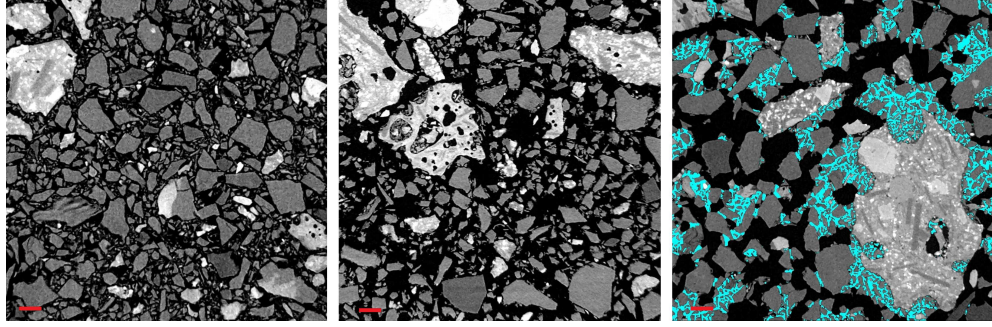


Fig. 10: 2D Micro-CT reconstructions of cemented samples containing 1, 3 and 12 wt.% water content (left to right) imaged under ambient conditions prior to freezing. Modified from [Ricardo et al. \(2026\)](#) with 100 μm scale bar in red, water in cyan, lithics in grey and void in black.

at water content of 1, 3 and 12 wt.%. Increasing water content can be seen to have two effects, first it coats grains surfaces and fully encapsulates grains in a liquid medium, and second it coalesces loose, individual lithic grains into clusters, with mean pore size increasing from 0.057 ± 0.018 mm to 0.062 ± 0.022 ([Ricardo et al. 2026](#)). It is well known that the addition of liquid water corresponds to an increase in the tensile strength of terrestrial soils as a function of soil water characteristic curve (SWCC) ([Yin and Vanapalli 2018](#)). So when frozen, this interconnected network of water that coats and binds grains together becomes a cement that reinforces the entire granular skeleton ([Atkinson et al. 2020](#)). In conclusion, we show clear quantitative evidence the observed change in G_{Lunar} values over several orders of magnitude for ice-cemented samples is due to differences in soil saturation regime, a process not observed in any other icy regolith sample.

4.2 Granular icy regolith

Unsintered and pressure sintered icy regolith exhibited lower sensitivity to changes in geotechnical behavior, attributed to the absence of structural changes in the granular skeleton with increasing ice content. Instead, ice occurs as discrete grains suspended in a groundmass of lithic particles ([Ricardo et al. 2026](#); [Purrrington et al. 2022](#); [Sowers et al. 2019](#); [Johnson et al. 2025](#)), where geotechnical behavior of the material is governed principally by interparticle friction and the volume filling fraction of ice grains ([Haack et al. 2020](#); [Brisset et al. 2022](#)). Smaller, angular grains have higher coordination numbers (average number of particle contacts) resulting in greater interparticle friction and cohesive forces ([Schellart 2000](#)). This explains why ice content and strength are positively correlated for CI (Figure 4) G_{Lunar} (Figure 5) and $c - \phi$ curves (Figure 7). Increased bulk density and the formation of sinter-necks between grains increases strength further, as external forces acting on the granular skeleton (i.e. cone tip) must overcome the tensile-strength of the sinter-neck, as well as large sintered ice-aggregates having reduced ability to translate and rotate, due to higher interparticle

friction and compaction (Gundlach et al. 2018; Thomas et al. 1994). This was experimentally observed as larger sawtooth fluctuations in the penetration curves of sintered versus unsintered samples (Figures A4 and A3), indicating the sequential fracturing of structures (i.e. build-up then release of load) beneath the advancing cone tip. The formation of sinter necks occur via mass transport from the ice-grain to the contact neck via plastic flow, sublimation-deposition, volume and surface diffusion (Blackford 2007), and is driven by the need of the system to minimize its surface energy. This process is a direct consequence of the Laplace and Kelvin equations and is the basis of Ostwald ripening (Gundlach et al. 2018; Blackford 2007; Bartels-Rausch et al. 2012). A process where large grains grow at expense of smaller grains, as observed by Jost et al. (2016, Fig. 5).

For comparison, Johnson et al. (2023) sinter samples with 0 to 10 wt.% ice using isostatic loads of 0.797 MPa within a walk-in freezer held at approximately -10°C to -20°C (263 to 253 K) resulting in maximum CI values ranging from <1 to >13 MPa. Here, all samples were sintered under higher isostatic loads ($1.097^{+0.058}_{-0.081}$ MPa) but at lower temperature ($-180 \pm 5^{\circ}\text{C}$ or 93 K) resulting in all measurements to have maximum CI values below approximately 0.8 MPa (Table A1). Because the sintering rate, and therefore size of the sinter neck and its strength, is temperature dependant (Gundlach et al. 2018) the disparity with results from Johnson et al. (2023) is attributed to differences in temperature during sample preparation. In other words, colder temperatures correspond to less sintering and therefore lower strength.

4.3 Vapor deposited icy regolith

The vapor deposited specimen containing 6.68 wt.% ice exhibited intermediate strength between that of ice-cemented and pressure sintered morphologies. The $c-\phi$ analysis suggest it had higher cohesion than measurements at the base of the Apollo 15 trench, suggesting that if vapor deposited icy regolith exists on the Moon it may be as strong as deep, compacted dry regolith. G_{Lunar} values of the vapor deposited sample were between 66% lower or 23% higher compared to the upper limit of Apollo 16 measurements, depending if G_{Lunar} was calculated over the full depth (30 mm) or to the peak of the ice lens strength (15 mm). To better understand how the strength of vapor deposited samples changes with increasing ice content, future work covering a wider range of sample ice content and bulk densities are required.

Thermal pumping has been proposed as a mechanism for water vapor to migrate into the subsurface and form an intergranular vapor deposited ice fabric (Schorghofer and Aharonson 2014). However, conditions for strong pumping are estimated to be only 0.8% of the global surface area of the Moon (about 0.4% in each lunar polar region), with the total area where any pumping can occur estimated to be 1.1%. For these locations, pore-filling ice may occur in the upper tens of centimeters. Figure A6 shows the presence of a subsurface ice lens -interpreted by the rise then fall of penetration resistance- as would be expected for a thermally pumped subsurface ice lens. For regions where thermal pumping is ineffective (i.e. vapor cannot be thermally pumped downwards), vapor deposition will instead occur directly onto surface regolith, like a frost, which may correspond to observed surface ice deposits in PSRs (Li et al. 2018). These deposits would undergo space weathering and become buried and preserved by

ejecta (Cannon et al. 2020), and/or comminuted by micrometeorite impacts and diffusively mixed into the surface such as the 'Field Debris' model by Farrell et al. (2019). Like agglutinate formation, vaporized volatiles could redeposit onto lithic grains and form an iceglutinate, as illustrated in Johnson et al. (2023, Fig.5) and Ricardo et al. (2023, Fig. 20). Depending on the temperature of deposition, these ices would be partially crystalline or amorphous (Chaplin 2019), with amorphous ice being weaker and more porous (Bar-Nun and Laufer 2003). Knowledge of the polymorphic structure of lunar ice is poorly understood, but is expected to significantly impact thermal and mechanical extraction techniques.

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977 4.4 Excavation and trafficability of icy lunar regolith

978 Excavation efficiency was found to diverge dramatically, depending on ice morphology. 979 The energy required to excavate H_2O from pressure sintered samples became more 980 efficient at higher ice content, decreasing from 0.35 J/g of H_2O yield at 1 wt.% to 981 0.11 J/g at 30 wt.% - a 3.2-fold improvement. Conversely, excavation of ice-cemented 982 samples became less efficient, increasing from 2.74 J/g at 1 wt.% to 143.03 J/g at 12 983 wt.% - a 52-fold reduction in efficiency. These relationships (given by Equations 4 and 984 3) indicate that ice enriched (e.g 30 wt.%) deposits of pressure sintered icy regolith are 985 more valuable than lower grade deposits, whilst the opposite is true if one uses only 986 ice-cemented regolith to predict ice excavatability on the Moon. This highlights how 987 the choice of icy simulant used in experimental studies has fundamental implications 988 on site selection and is a non-trivial risk for ISRU planning.

989 To predict trafficability in lunar polar regions, experimental data obtained here is 990 compared to *in-situ* observations from the Apollo 16 SRP and Lunar Roving Vehicle 991 (LRV). G values derived from the Apollo 16 SRP over the full (20 to 70 cm) and 992 modified (3 cm) depth were 0.42 kPa mm⁻¹ to 4.13 kPa mm⁻¹ and 6.85 kPa mm⁻¹ to 993 37.33 kPa mm⁻¹ represented by the shaded green and yellow regions in Figure 6 respec- 994 tively. Assuming that equation A6 provides a reasonable estimate for the change in 995 regolith strength with density, and that G_{Lunar} is a reasonable proxy for trafficability 996 (Daca et al. 2021; Daca and Skonieczny 2022), then the addition of ice increases traf- 997 ficability across all tested conditions, independent of ice content, morphology, or bulk 998 density. Vapor deposited and pressure sintered samples containing >5 wt.% ice had 999 G_{Lunar} values between 7.79 kPa mm⁻¹ to 69.75 kPa mm⁻¹, which fall within or slightly 1000 above the range of modified Apollo 16 G values (6.85 kPa mm⁻¹ to 37.33 kPa mm⁻¹), 1001 indicating that these types of ice deposits could have similar to slightly higher traffica- 1002 bility than dry Highlands regolith. At 1 wt.% ice, the G_{Lunar} of pressure sintered ice 1003 was 5.54 kPa mm⁻¹, a reduction of only 19%. The vapor deposited sample with an ice 1004 content of 6.68 wt.% and a bulk density of 1.5 g cm⁻³ had G_{Lunar} values 22.57% higher 1005 than the upper limit of modified Apollo 16 measurements, and was 292% stronger 1006 than a pressure sintered sample containing 8 wt.% ice at 1.67 g cm⁻³ density. These 1007 results suggest that while the presence of ice enhances trafficability, the magnitude 1008 of this improvement depends strongly on ice morphology, as measured using G slope 1009 parameter.

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5 Conclusions

A series of cone penetration tests were performed with sample temperatures of approximately 90 K, initial probe temperatures between 90 K to 113 K, and ice contents ranging from 0 to 30%. The ice microstructure of each sample included; i) ice-cemented frozen ‘mud-pies’, ii) unsintered granular icy regolith, iii) pressure sintered granular icy regolith, and iv) vapor deposited icy regolith.

The experimental results presented here confirm that the geotechnical behaviour of icy lunar regolith simulants are primarily a function of ice microstructure, which depends on the physical state of water during sample preparation. Whilst the addition of any ice regardless of its microstructure resulted in elevated strength compared to dry regolith, the nature and extent of this increase varied by several orders of magnitude. This has direct implications for predicting the performance of trafficability and excavatability within PSRs using different icy simulants. The key findings of this work are as follows.

1. The addition of any ice, regardless of its microstructure, resulted in elevated strength compared to dry regolith measured with CI (Figure 4) and trafficability measured using the G slope parameter (Figure 6) (Daca et al. 2021; Daca and Skonieczny 2022). The magnitude of this improvement depends primarily on ice morphology.
2. The maximum CI for dry LHS-1 compacted to 1.55 g cm^{-3} was $33.63 \pm 8.69 \text{ kPa}$. The addition of only 1 wt.% ice increased CI by 74%, 1180%, and 6077% for unsintered, pressure sintered and ice-cemented icy simulants respectively, a difference of nearly 2 orders of magnitude (Figure 4). At 3 wt.% ice, this difference increased to more than 2.5 orders of magnitude despite having similar sample density, temperatures and being tested under identical conditions.
3. After correcting for gravity, the measured G_{Lunar} for all icy regolith samples was higher than predicted values for dry regolith (Figure 6) determined using the equation :
$$G_{lunar} = 6.1 \times 10^{-8} e^{10.524\rho_b} \cdot (-1.65\rho_b + 3.25) \quad (5)$$
4. In all metrics (CI , G_{Lunar} , $C - \phi$, ϵ , k , τ) there is a clear disparity between ice-cemented and all other icy regolith morphologies, as well as *in-situ* measurements of lunar regolith with the Apollo SRP. This contributes to growing evidence that the use of ice-cemented simulants or concrete and rock analogues may be flawed, if ice in lunar regolith occurs as different morphologies. For example, despite having four times more ice and nearly 15% higher bulk density (1.6 g cm^{-3}), a pressure sintered sample containing 12 wt.% ice had CI values 97.6% (42-times) lower than an ice-cemented sample with 3 wt.% ice and density of 1.5 g cm^{-3} .
5. The choice of icy simulant in a study has fundamental implications on site selection and is a non-trivial risk for ISRU planning. For example, excavation efficiency as a function of ice content for pressure sintered samples increased 3.2-fold, from 0.35 J/g of H_2O yield at 1 wt.% to 0.11 J/g at 30 wt.%, given by the equation:

$$\eta_{exc}(wt_{ice}) = -4.13wt_{ice}^2 + 67.9wt_{ice} - 77.8 \quad (6)$$

whilst it decreased 52-fold for ice cemented samples, increasing from 2.74 J/g at 1 wt.% to 143.03 J/g at 12 wt.% given by the equation:

$$\eta_{exc}(wt_{ice}) = \frac{0.313}{wt_{ice}} + 0.0356 \quad (7)$$

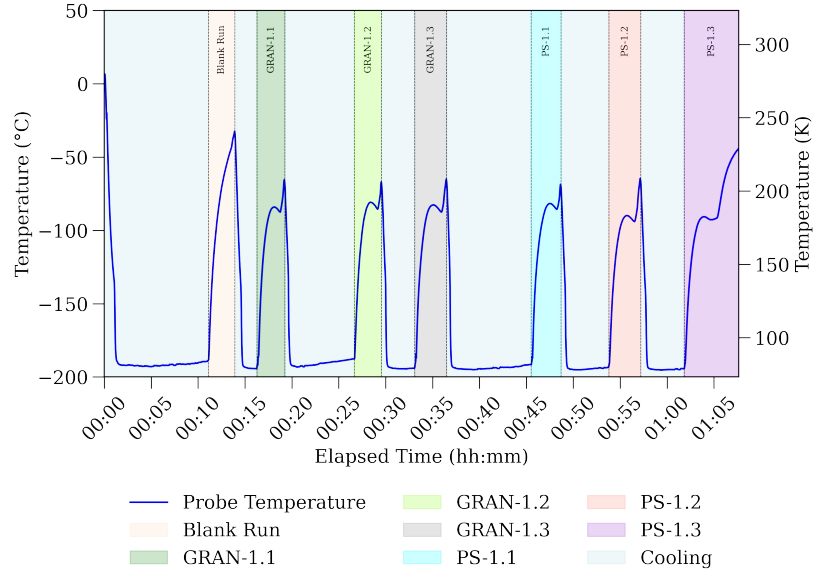
6. The vapor deposited sample with an ice content of 6.68 wt.% and a bulk density of 1.5 g cm^{-3} exhibited high strength, being 85% stronger than the Apollo 16 SRP lower limit (6.85 kPa mm^{-1}) even after correcting SRP data to 30 mm depth, and became 22.57% higher than the Apollo 16 upper limit ($37.33 \text{ kPa mm}^{-1}$) after recalculating G_{Lunar} to the depth of the observed ice lens ($\approx 15 \text{ mm}$ in Figure A6). This indicates that buried, unweathered (i.e. unfractured) lenses of vapor deposited ice in lunar PSRs may provide similar to slightly improved trafficability to dry highlands regolith explored during Apollo 16. This assumes the upper 30 mm has commensurate density. However it is known that the upper surface of PSRs may be more porous than illuminated regions, which will result in poorer traction and increased wheel slip (Byron et al. 2019; Jordan et al. 2015).

This work acknowledges and is intended to build upon a growing body of experimental literature focused on evaluating the geotechnical properties of icy lunar regolith simulants (Atkinson et al. 2020; Gertsch et al. 2008; Johnson et al. 2023). Novel contributions include the first quantitative geotechnical measurements of vapor deposited icy lunar regolith, as well as measurements of the specific energy and excavation efficiency for pressure sintered icy simulants. Results presented here provide clear quantitative evidence that ice-cemented samples exhibit markedly different geotechnical behaviour compared to granular or vapor deposited morphologies tested under similar conditions. This aligns with a similar call to action by Johnson (2024b) and contributes to a growing body of work that suggest alternative morphologies of icy regolith which better emulate ice formation on the Moon should be used (Johnson et al. 2023; Johnson 2024a; Sowers et al. 2019; Purrington et al. 2022). For this reason, vapor deposited ice and granular icy simulants prepared under cryogenic conditions ($\leq 110 \text{ K}$) are recommended for use in large scale (e.g. regolith test bin) studies of trafficability or excavatability.

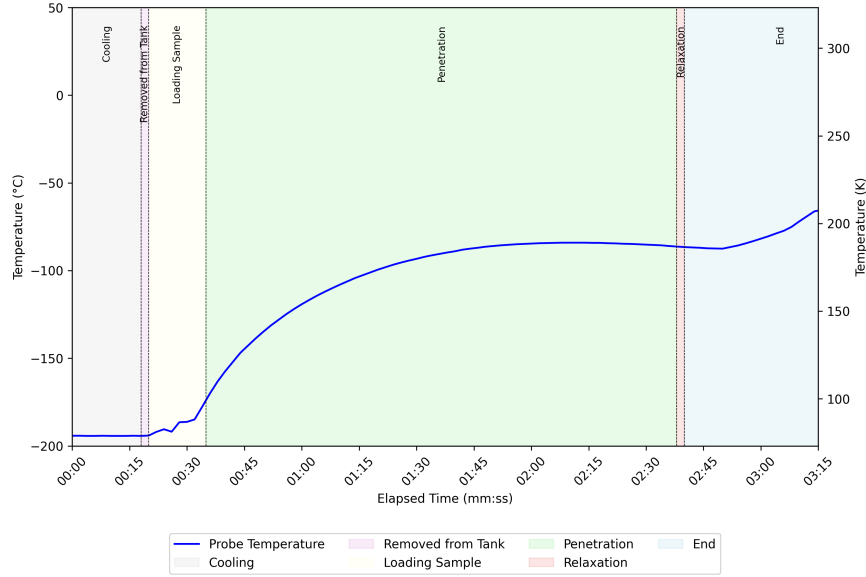
Supplementary Information. Supplementary figures and datasets are available online as Online Resources.

Acknowledgements. This work was supported by the ANFF-VIC Biointerface Engineering Hub, part of the Victorian Node of the Australian National Fabrication Facility (ANFF), a company established under the National Collaborative Research Infrastructure Strategy to provide nano and microfabrication facilities for Australia’s researchers. The authors also acknowledge the financial contributions and support of the CSIRO Mineral Resources group, the Commonwealth Research Training Program Scholarship, and the CSIRO research fellowship program.

Declarations	1105
	1106
Author Contributions. D.R developed the conceptual framework, conducted all experiments, prepared figures, collated data and wrote the manuscript. J.H, M.A.R and G.B reviewed the manuscript and supervised this study. All authors reviewed and approved the final version of the manuscript.	1107
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Data Availability. Full datasets and code used to generate all experimental tables and figures here are available upon request by contacting the corresponding author.	1111
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Ethics approval. Ethics and Consent to Participate declarations are not applicable.	1114
	1115
Appendix A Appendices	1116
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A.1 Experimental Controls	1118
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(a)



(b)

Fig. A1: (Top) Change in probe temperature with time for a series of granular (GRAN-1) and pressure sintered (PS-1) regolith simulant containing 1 wt.% ice content measured in triplicate. (Bottom) Characteristic change in probe temperature with time. Gradual increase (0:30 to 1:45) due to exposure to ambient laboratory conditions. Gradual decrease (1:45 to 2:45) due to probe insertion into cryogenic sample, with subsequent increase due to gradual warming under atmospheric conditions.

A.2 Calculations

A.2.1 Cone Index and Cone Index Gradient

The cone index, CI, represents the force per unit base area required to push the cone into soil. It is a measure of soil strength at specific depths and is a combination of soil properties including shear strength, compressability and sidewall friction (Shoop 1993; Mulqueen et al. 1977). CI is used interchangeably with cone tip resistance, q_c , which is used more widely in modern terrestrial geotechnical surveying (Lunne 1997; Robertson 1990; Robertson and Cabal 2022) and is reported in kilopascals (1 kPa = 1 kN m⁻²) calculated following the equation given in ASTM D5778-20 (ASTM 2014):

$$CI = q_c = \frac{Q_c}{A_c} \quad (\text{A1})$$

where Q_c is the total force acting on the cone tip (kN) and A_c is the projected circular area of the cone (m²). For the 6 mm diameter cone used in this study, A_c is 2.827×10^{-5} m². As CI is used in studies of lunar regolith simulants (Lucas et al. 2024) and was used to describe Apollo 15 and 16 SRP data (Mitchell et al. 1972b,a), CI rather than q_c was used to describe results here.

The cone index gradient, G (kPa/mm) represents the relative soil strength (kPa) over a range of depths (mm) and is calculated as the slope of the linear regression of the penetration verses depth curve using the equation (Oravec et al. 2008):

$$G = \frac{\sum_{i=1}^n (d_i - \bar{d})(CI_i - \overline{CI})}{\sum_{i=1}^n (d_i - \bar{d})^2} \quad (\text{A2})$$

where

$$\bar{d} = \frac{\sum_{i=1}^n d_i}{n} \quad \text{and} \quad \overline{CI} = \frac{\sum_{i=1}^n CI_i}{n} \quad (\text{A3})$$

n is the number of measurements in an insertion, i is the measurement number, d_i and CI_i are the depth and cone index at point i respectively, and $\bar{d}$ and $\overline{CI}$ are the arithmetical mean of all the depth and cone index values measured in mm or kPa respectively (Mcrae et al. 1965; Daca et al. 2021). After calculating G for each individual sample using Equation A2, the mean ($\bar{G}$) and standard deviation (σ_G) for a group of samples with identical ice content and microstructure was calculated as:

$$\bar{G} = \frac{1}{n} \sum_{i=1}^n G_i \quad \text{and} \quad \sigma_G = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (G_i - \bar{G})^2} \quad (\text{A4})$$

where n is the number of identical samples, and G_i is the cone index gradient of the i -th sample. As the error in depth measurements obtained here using the UTM was essentially negligible ($\pm < 0.001$ mm), we do not propagate error in both G and depth measurements, as was done by Lucas et al. (2024) who have depth errors of ± 1 mm due to use of handheld Rimik CP40II electronic cone penetrometer which has lower accuracy.

1243 A.2.2 Correcting for lunar gravity

1244 To convert G and ρ_b into G_{Lunar} a gravity reduction factor, R_f from Lucas et al.
1245 (2024) was used. Because R_f is calculated using ρ_b as:

$$1247 R_f = -1.65\rho_b + 3.25 \quad (A5)$$

1248 and Lucas et al. (2024) established an empirical relationship between G and ρ_b , the
1249 predicted variation of G_{lunar} with bulk density can be expressed as:

$$1252 G_{lunar} = 6.1 \times 10^{-8} e^{10.524\rho_b} \cdot (-1.65\rho_b + 3.25) \quad (A6)$$

1254 A.2.3 In-Situ Determination of Engineering Properties

1255 Literature values for the geotechnical properties of lunar soil like relative density
1256 (R_d), porosity (n), cohesion (c), and soil friction angle (ϕ) are based on measurements
1257 obtained by the Apollo SRP (Mitchell et al. 1972a,b, 1974). These measurements were
1258 derived using equations by Durgunoglu and Mitchell (1973), who model the failure
1259 envelope of a cone penetrometer by modifying Terzaghi's theory of bearing capacity
1260 for circular footings (i.e wedge and cone penetrometers) using circular shape factors
1261 for cohesion (ξ_c) and surcharge (ξ_γ), and by combining the bearing capacity factors
1262 for surcharge (N_γ) and friction (N_q), calculated using the equation:

$$1264 q_f = cN_c\xi_c + \gamma_s B N_{\gamma q} \xi_{\gamma q} \quad (A7)$$

1265 where q_f is the end bearing capacity or unit penetration resistance (N cm^{-2}), c is soil
1266 cohesion, (N cm^{-2}), γ_s is unit soil weight (N cm^{-3}), B is penetrometer width (cm),
1267 and N_c and $N_{\gamma q}$ represent the cohesion and friction–surcharge bearing capacity factors
1268 respectively. Durgunoglu and Mitchell (1973) also outline a procedure for presenting
1269 plots of c as a function of ϕ for given values of q_f and B . These cohesion-friction ($c-\phi$)
1270 curves can be obtained given known values of cone semi-apex angle (α , °), the ratio
1271 of cone friction (δ) to soil friction (ϕ) given as δ/ϕ , and the ratio of cone depth (D)
1272 relative to cone diameter (B) given as D/B .

1273 The soil bearing capacity factors for cohesion (N_c) and friction-surcharge ($N_{\gamma q}$)
1274 were calculated using the equations (Durgunoglu and Mitchell 1973, p. 191,202):

$$1277 N_c = \frac{[1 + \sin \phi \sin(2\gamma - \phi)] \cdot e^{2\theta_o} \tan \phi}{\sin \phi \cdot \cos \phi} - \frac{1}{\tan \phi} \quad (A8)$$

$$1280 + \frac{\cos(2\gamma - \phi) \cdot \tan \psi \cdot e^{2\theta_o} \tan \phi}{\cos \phi}$$

1282 and

$$1284 N_{\gamma q} = \left[\frac{\cos(\psi - \delta)}{\cos \delta} \right] \cdot \left[\frac{1 + \sin \phi \sin(2\gamma - \phi)}{\cos \phi \cos(\gamma - \phi)} \right]$$

$$1286 \cdot \left[\frac{1}{4} \frac{\cos^2(\gamma - \phi)}{\cos^2 \psi \cos^2 \phi} I_\theta \right]$$

$$\begin{aligned}
& + \frac{3 \cos(\gamma - \phi)}{4 \cos \psi \cos \phi} \cos^2 \beta e^{2\theta_0 \tan \phi} \left(m - \frac{2}{3} m' \right) \\
& + K \frac{\cos \psi \cos \phi}{\cos(\gamma - \phi)} m^3 \\
& - K_0 \frac{\cos \psi \cos \phi}{\cos(\gamma - \phi)} (m - m')^2 (m + 2m') \left] - \frac{\tan \psi}{4} \right. \quad (A9)
\end{aligned}$$

where

q_f = unit penetration resistance (N cm^{-2})	1289
c = soil cohesion (N cm^{-2})	1290
γ_s = unit soil weight (N cm^{-3})	1291
N_c = bearing capacity cohesion factor	1292
$N_{\gamma q}$ = bearing capacity friction-surcharge factor	1293
ξ_c = cohesion shape factor	1294
$\xi_{\gamma q}$ = friction-surcharge shape factor	1295
B = width of penetrometer base (cm)	1296
ϕ = angle of internal friction of soil ($^\circ$)	1297
$\psi = 90^\circ - \alpha$ (α = semi-apex angle ($^\circ$))	1298
δ = penetrometer base to soil friction angle ($^\circ$)	1299
γ = the topmost angle of the plane shear zone ($^\circ$)	1300
$\theta_o = 180^\circ - (\psi + \gamma) + \beta$	1301
K = lateral earth pressure coefficient (assumed $K = K_0 = 1 - \sin \phi$)	1302
α = semi-apex angle of cone ($^\circ$)	1303
δ/ϕ = base roughness (0 (perfectly smooth) to 1 (perfectly rough))	1304
$D/B = m$ = Cone base Depth below surface (D) to diameter (B) ratio	1305
$m' = \frac{D_\beta}{B} = \frac{\cos(\gamma - \phi) \sin \beta}{2 \cos \psi \cos \phi} e^{\theta_o \tan \phi}$	1306

and I_θ was calculated as (Durgunoglu and Mitchell 1973, p. 195):

$$\begin{aligned}
I_\theta = \frac{1}{1 + 9 \tan^2 \phi} \{ & 3 \tan \phi [e^{3\theta_0 \tan \phi} \cos \beta - \cos(\theta_0 - \beta)] + \\
& [e^{3\theta_0 \tan \phi} \sin \beta + \sin(\theta_0 - \beta)] \} \quad (A10)
\end{aligned}$$

Shape factor values determined semi-empirically by Brinch and Hansen(1966) were experimentally validated by Durgunoglu and Mitchell (1973) to follow the form:

$$\xi_c = 1.0 + [(0.2 + \tan \phi)(B/L)] \quad (A11)$$

where ξ_c is the shape factor for cohesion, and $\xi_{\gamma q}$ is for the friction-surcharge term calculated as:

$$\xi_{\gamma q} = 1.0 - 0.4(B/L) + \frac{1.5}{\frac{B}{D} + \frac{1.5}{(0.6 + \tan^6 \phi)(B/L)}} \quad (\text{A12})$$

where B = penetrometer width and L is length of footing base. For a circular (i.e cone) penetrometer $B = L$ and so equation A12 simplifies to:

$$\xi_{\gamma q} = 0.6 + \frac{1.5}{\frac{B}{D} + \frac{1.5}{0.6 \cdot \tan(\phi)^6}} \quad (\text{A13})$$

For soils with both cohesive, c and frictional, ϕ there are an infinite combination of c and ϕ values which satisfy equation A7. Durgunoglu and Mitchell (1973) outline a procedure for presenting plots of c as a function of ϕ required to give the measured penetration resistance for a given penetration depth. These curves of cohesion-friction, $c - \phi$, given values of α , δ/ϕ and D/B , outlined below.

1. Assume initial value of ϕ
2. For given α , δ/ϕ and D/B , iteratively calculate values of γ and β using procedures outlined using Fortran code in Appendices of Durgunoglu and Mitchell (1973), which are also given in Python and schematically shown in Ricardo (2025).
3. Given γ and β , determine corresponding values of N_c and $N_{\gamma q}$ using equations A8 and A9, where I_θ is determined using equation A10
4. Determine values for shape factors ξ_c and $\xi_{\gamma q}$ for assumed ϕ and known D/B using equations A11 and A13
5. Substitute vales of N_c , $N_{\gamma q}$, ξ_c and $\xi_{\gamma q}$ from step 2 and 3 into equation A7 and solve for c with known value of cone bearing capacity, q_f
6. Plot resulting $c - \phi$ relationship

A.2.4 Specific Energy and Excavation Efficiency

The SE of displacement for all samples was calculated from the volume of material displaced by the cone penetrometer (V_{disp} , cm^3) as a function of maximum penetration depth (d) and the work required to achieve this depth. For penetration depths less than or equal to the height of the cone tip (h_{cone}), the volume was calculated as the volume of a cone. For penetration depths greater than h_{cone} , the total displaced volume was calculated as the sum of the cone tip volume and the cylindrical rod volume using the equation:

$$V_{\text{disp}}(d) = \begin{cases} \frac{1}{3}\pi r_{\text{cone}}^2 d, & \text{if } d \leq h_{\text{cone}} \\ \frac{1}{3}\pi r_{\text{cone}}^2 h_{\text{cone}} + \pi r_{\text{rod}}^2 (d - h_{\text{cone}}), & \text{if } d > h_{\text{cone}} \end{cases} \quad (\text{A14})$$

where for a 6 mm diameter cone with an cone apex of 60° the height of the cone (h_{cone}) is 5.2 mm, and the radius of the cone (r_{cone}) and rod (r_{rod}) are 3 mm.

Work was calculated as the area under the load (N) versus displacement (mm) curve using the trapezoidal rule by summing the areas of trapezoids formed between

consecutive data points using the equation:

$$\text{Area} \approx \int_a^b f(x) dx \approx \sum_{i=1}^{n-1} \frac{(x_{i+1} - x_i)}{2} [f(x_i) + f(x_{i+1})] \quad (\text{A15})$$

where x_i is the displacement values (mm), $f(x_i)$ are the load values (N), and n are the number of data points.

To quantify the energy required to excavate a unit mass of water from icy lunar deposits and to consider the enrichment of ice content for different deposits, excavation efficiency (η_{exc} , J/g) was calculated as the ratio of the work done during penetration to the mass of water (m_{water} , g) that was excavated using the equation :

$$\eta_{\text{exc}} = \frac{W}{m_{\text{water}}} \quad (\text{A16})$$

where m_{water} is calculated using the equation:

$$m_{\text{water}} = m_{\text{exc}} \cdot \left(\frac{wt_{\text{ice}}}{100} \right) \quad (\text{A17})$$

using known values of excavated mass (m_{exc} , g) and sample ice content (wt_{ice} , %) from Table A4 and A1 respectively. η_{exc} can then be used to evaluate which ice deposits would yield the highest mass of water, per unit energy. The influence of ice content and microstructure on the mean and standard deviation of the SE of displacement, SE of excavation and η_{exc} are provided in Figure 8, together with the excavatability metrics calculated here, listed in Table A4.

A.2.5 Modelling Relaxation

In addition to characterizing various metrics associated with cone penetration to characterize geotechnical behaviour, a growing body of literature (Atkinson et al. 2019, 2020; Thrift and Dreyer 2022; Thrift et al. 2023; Thrift and Dreyer 2025) has shown that modelling relaxation behaviour can provide insights on elasticity and viscous elements for icy regolith simulants. The Maxwell model is a common rheological framework for describing material relaxation, comprising an elastic spring in series with a viscous dashpot to capture both instantaneous (elastic) and time-dependent (viscous) deformation (Liingaard et al. 2004; Atkinson 2023). In generalized form (Peleg and Normand 1983), multiple Maxwell elements can be arranged in parallel to represent complex viscoelastic behavior. Atkinson et al. (2019) found that a two-arm model best fit the relaxation behavior of lunar regolith simulant JSC-1A studied using cone penetrometer, given by the equation:

$$\frac{\sigma(t)}{\sigma_{\text{max}}} = \epsilon(k_e + k_1 e^{-\frac{t}{\tau_1}} + k_2 e^{-\frac{t}{\tau_2}}) \quad (\text{A18})$$

where $\sigma(t)$ is the vertical stress exerted on the probe tip as a function time (Pa), σ_{max} is the maximum resistance experienced by the probe, ϵ is the strain experienced by

1427 the sample after penetration, k_e is the residual load supported by the material after
1428 relaxation has occurred (Pa), and the elastic and viscous time-dependent components
1429 of the i^{th} Maxwell arm are represented by k_i and τ_i (S) respectively.

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A.3 Extended Dataset

A.3.1 Cone Index with Depth

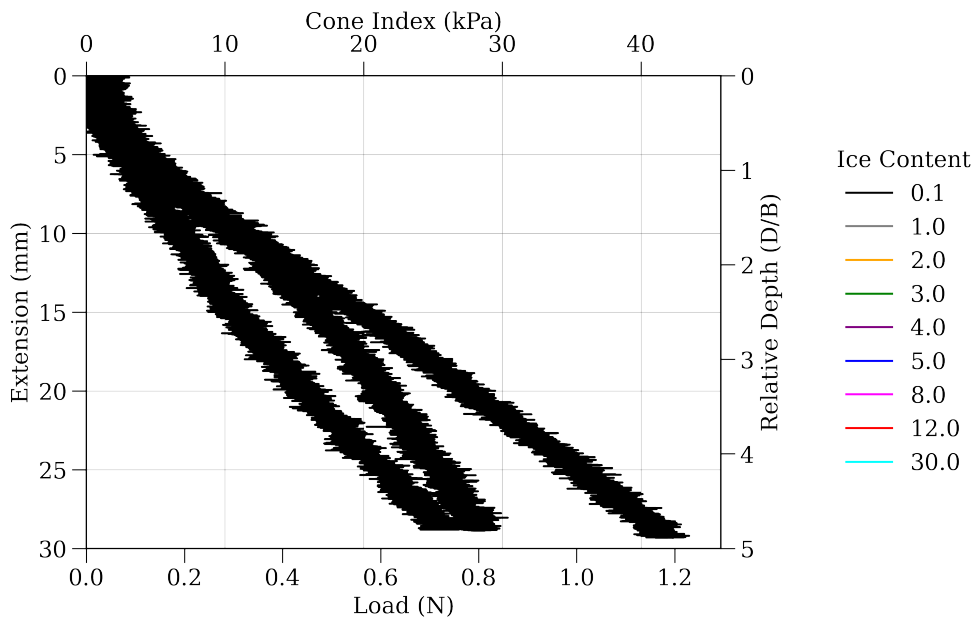


Fig. A2: Cone Index with depth for dry LHS-1

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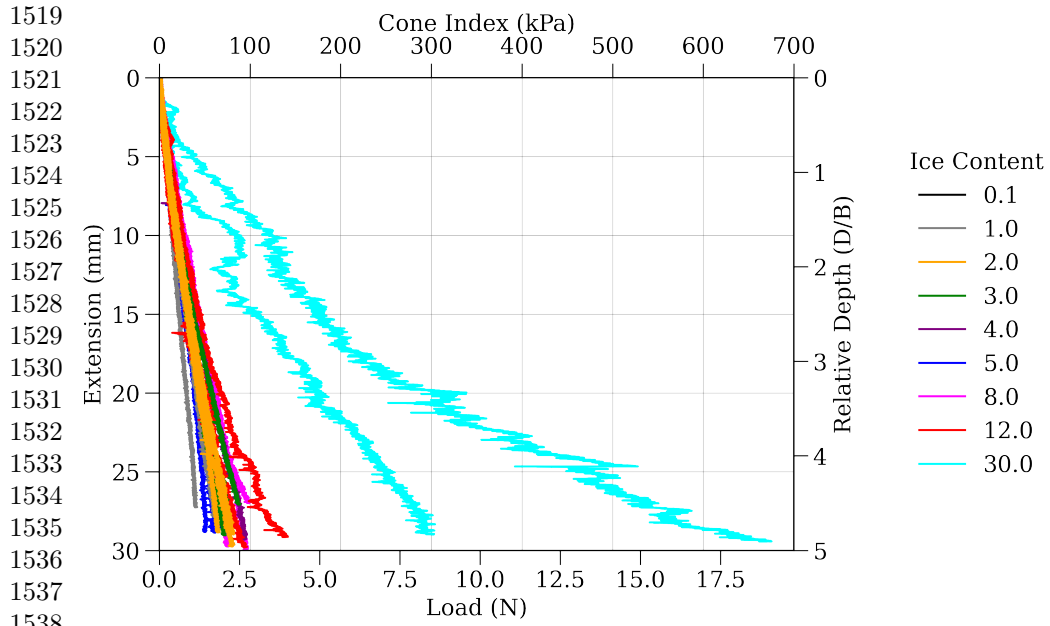


Fig. A3: Cone Index with depth for unfused granular icy regolith simulant

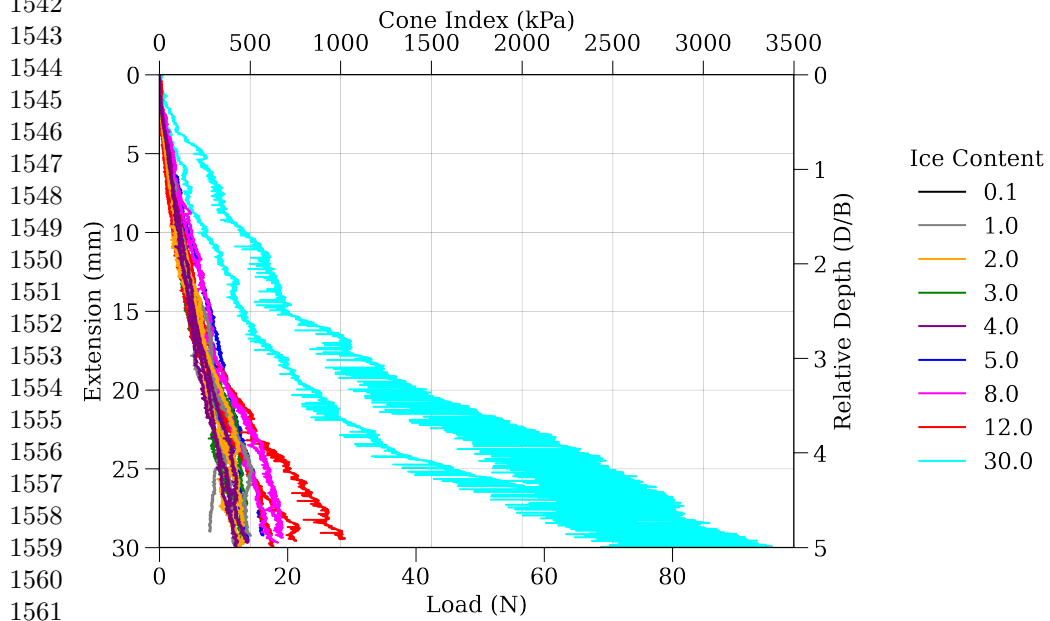


Fig. A4: Cone Index with depth for pressure sintered icy regolith simulant

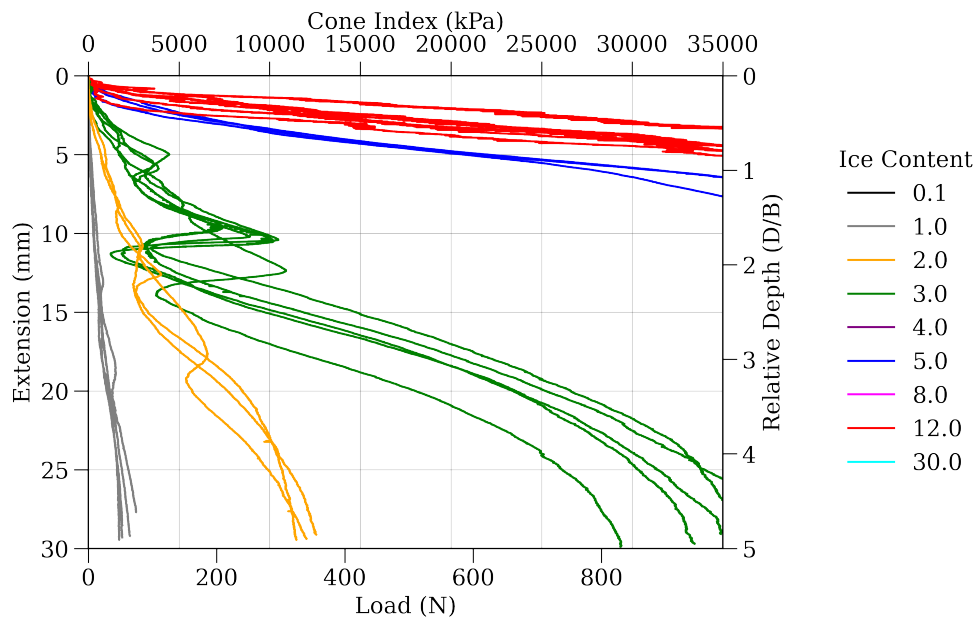


Fig. A5: Cone Index with depth for ice-cemented regolith simulant

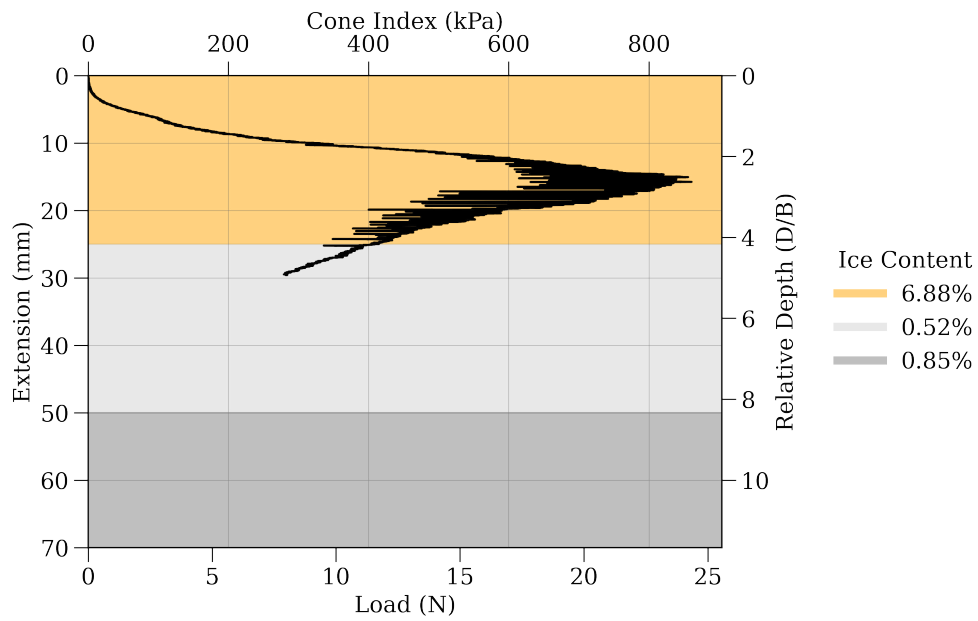


Fig. A6: Cone Index with depth for vapor deposited icy regolith simulant

1611 **A.3.2** Sample Properties
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1613 **A.3.3** Correcting G for Lunar gravity
1614 **A.3.4** Specific Energy
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Table A1: Summary of dry and icy lunar regolith simulant characteristics. Sample identifiers denote simulant type: dry (DRY), pressure sintered (PS), granular (GRAN), and ice-cemented (CEM). Associated number indicates target ice content(e.g., PS-3 = 3 wt.%). Standard deviation of 0.0 indicate error smaller than 2 decimal places.

Identifier	Ice Content (wt_{ice} , %)	Bulk Density ρ_b (g/cm^3)	Porosity n (%)	Max Load (N)	Max Cone Index (kPa)	Cone Index Gradient G (kPa/mm)
DRY	0.1 ± 0.0	1.55 ± 0.02	44.66 ± 0.65	0.95 ± 0.25	33.63 ± 8.69	1.16 ± 0.34
GRAN-1	1.03 ± 0.03	1.5 ± 0.0	46.25 ± 0.09	1.65 ± 0.42	58.45 ± 14.9	2.13 ± 0.58
GRAN-2	2.01 ± 0.02	1.55 ± 0.04	44.15 ± 1.2	2.19 ± 0.26	77.37 ± 9.35	2.64 ± 0.21
GRAN-3	3.03 ± 0.05	1.5 ± 0.0	45.54 ± 0.08	2.26 ± 0.24	80.03 ± 8.49	2.84 ± 0.44
GRAN-4	3.99 ± 0.02	1.51 ± 0.02	44.82 ± 0.71	2.24 ± 0.43	79.26 ± 15.32	2.8 ± 0.55
GRAN-5	4.97 ± 0.06	1.5 ± 0.0	44.76 ± 0.02	1.77 ± 0.25	62.72 ± 8.99	2.24 ± 0.24
GRAN-8	8.0 ± 0.02	1.5 ± 0.0	43.59 ± 0.11	2.6 ± 0.36	92.04 ± 12.65	3.08 ± 0.45
GRAN-12	12.01 ± 0.0	1.5 ± 0.0	41.87 ± 0.04	3.04 ± 0.86	107.57 ± 30.35	3.4 ± 0.91
GRAN-30	30.0 ± 0.0	1.38 ± 0.23	38.45 ± 10.08	13.85 ± 7.42	489.86 ± 262.5	15.97 ± 7.04
CEM-1	0.99 ± 0.01	1.5 ± 0.0	46.25 ± 0.05	58.74 ± 10.75	2077.64 ± 380.33	78.73 ± 10.46
CEM-2	2.0 ± 0.01	1.5 ± 0.01	45.81 ± 0.11	340.21 ± 15.27	12032.29 ± 540.03	462.95 ± 25.98
CEM-3	2.99 ± 0.01	1.5 ± 0.0	45.52 ± 0.05	$953.06^1 \pm 71.5$	33707.68 ± 2528.92	1377.8 ± 168.42
CEM-5	4.99 ± 0.01	1.5 ± 0.0	44.76 ± 0.06	$1000.9^1 \pm 0.41$	35399.66 ± 14.33	5540.07 ± 275.4
CEM-12	11.94 ± 0.08	1.5 ± 0.0	42.18 ± 0.1	$1002.57^1 \pm 1.98$	35458.54 ± 70.03	8878.35 ± 1597.82
PS-1	1.03 ± 0.03	1.76 ± 0.01	36.94 ± 0.46	12.17 ± 2.28	430.35 ± 80.47	15.99 ± 3.55
PS-2	2.01 ± 0.02	1.75 ± 0.0	36.94 ± 0.04	12.86 ± 0.38	454.69 ± 13.44	16.89 ± 0.35
PS-3	3.03 ± 0.05	1.72 ± 0.01	37.48 ± 0.35	12.99 ± 0.77	459.58 ± 27.3	17.08 ± 1.63
PS-4	3.99 ± 0.02	1.72 ± 0.02	37.24 ± 0.46	12.66 ± 1.14	447.62 ± 40.2	15.42 ± 1.52
PS-5	4.97 ± 0.06	1.72 ± 0.02	36.67 ± 0.76	14.47 ± 1.59	511.82 ± 56.1	18.9 ± 2.38
PS-8	8.0 ± 0.02	1.67 ± 0.0	37.15 ± 0.15	18.99 ± 0.42	671.7 ± 15.01	23.63 ± 1.47
PS-12	12.0 ± 0.0	1.6 ± 0.01	38.19 ± 0.15	22.89 ± 5.65	809.74 ± 199.97	26.2 ± 5.52
PS-30	30.0 ± 0.0	1.52 ± 0.25	32.1 ± 10.92	91.62 ± 5.52	3240.4 ± 195.15	94.46 ± 16.21
VAPOR	6.88	1.5	44.02	24.35	861.08	$16.39 (58.65)^2$

^aTest ended prematurely as the recorded load exceeded the 1 kN load cell test capacity

^bModified G calculated from the initial ramp of CI values to the depth of the observed ice lens (approximately 15 mm) in vapor deposited sample, seen in

Figure A6
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Table A2: Cone Index Gradient and associated error (%) for lunar regolith measured during Apollo 15 and Apollo 16 missions. Both full-depth and normalized 30 mm penetration analyses are shown. Data from the NASA Planetary Data System ([Mitchell et al. 2011, 2012](#)).

Apollo 15											
Station	Test	Index	Location	Cone Size (cm)	Max Load (N)	Max Stress (kPa)	Max Depth (cm)	Cone Index Gradient G (kPa/mm)	Error (%)	Modified G (kPa/mm) ¹	Error (%)
8	1	2	Adjacent to trench (North)	2.03	110.9	344	12.57	4.80	8.71	–	–
8	2	3	Bottom of trench	2.03	111.0	344	14.62	13.68	11.90	–	–
8	3	4	Rover tracks	2.03	110.3	342	10.05	5.07	16.32	–	–
8	4	5	Adjacent to rover	2.03	111.8	347	15.56	11.31	13.22	–	–
Apollo 16											
10	1	10	Near rover	2.03	219.3	680	21.9	2.36	21.75	29.32	33.2
10	2	11	Near rover	1.28	214.6	1663	50.4	3.61	13.71	37.33	53.0
10	3	12	Southwest of rover	1.28	214.2	1660	41.9	4.13	7.96	9.21	49.8
10	5	14	Southwest of rover	1.28	198.0	1535	62.6	1.55	22.65	28.07	30.0
4	1	5	Uphill south of rover	2.03	216.0	670	21.3	3.32	3.86	6.85	15.3
4	2	6	On bench, south of rover	1.28	149.5	429	74.2	0.42	22.30	–	–
4	3	7	On bench, southwest of rover	1.28	215.4	1669	45.4	2.67	17.35	–	–
4	4	8	Downhill, north of rover	1.28	221.6	1717	72.9	0.56	54.53	20.70	19.8

¹Modified G represents the cone index gradient computed from the first 30 mm of penetration data to better compare to experimentally derived data from icy regolith simulants which were done to 30 mm penetration depth.

Table A3: Summary of cone index gradient values of icy regolith simulants experimentally determined under terrestrial gravity (G_{Earth}), and corrected to lunar gravity (G_{Lunar}) using the density-dependent reduction factor R_f defined in Equation A5.

Identifier	Cone Index Gradient G_{Earth} (kPa/mm)	Density ρ (g/cm ³)	Reduction Factor R_f	Lunar-Corrected Cone Index Gradient G_{Lunar} (kPa/mm)
DRY	1.16 ± 0.34	1.55 ± 0.02	0.69	0.80 ± 0.23
GRAN-1	2.13 ± 0.58	1.50 ± 0.00	0.78	1.65 ± 0.45
GRAN-2	2.64 ± 0.21	1.55 ± 0.03	0.70	1.84 ± 0.15
GRAN-3	2.84 ± 0.44	1.50 ± 0.00	0.78	2.21 ± 0.34
GRAN-4	2.80 ± 0.55	1.51 ± 0.02	0.76	2.13 ± 0.42
GRAN-5	2.24 ± 0.24	1.50 ± 0.00	0.77	1.73 ± 0.19
GRAN-8	3.08 ± 0.45	1.50 ± 0.00	0.78	2.39 ± 0.35
GRAN-12	3.40 ± 0.91	1.50 ± 0.00	0.77	2.63 ± 0.71
GRAN-30	15.97 ± 7.04	1.38 ± 0.23	0.97	15.54 ± 6.85
CEM-1	78.73 ± 10.46	1.50 ± 0.00	0.77	60.91 ± 8.09
CEM-2	462.95 ± 25.98	1.50 ± 0.00	0.77	356.42 ± 20.00
CEM-3	1377.80 ± 168.42	1.50 ± 0.00	0.77	1064.52 ± 130.12
CEM-5	5540.07 ± 275.40	1.50 ± 0.00	0.77	4274.05 ± 212.46
CEM-12	8878.35 ± 1597.82	1.50 ± 0.00	0.78	6883.40 ± 1238.79
PS-1	15.99 ± 3.55	1.76 ± 0.01	0.35	5.54 ± 1.23
PS-2	16.89 ± 0.35	1.75 ± 0.00	0.37	6.18 ± 0.13
PS-3	17.08 ± 1.63	1.72 ± 0.01	0.41	7.01 ± 0.67
PS-4	15.42 ± 1.52	1.72 ± 0.01	0.42	6.45 ± 0.64
PS-5	18.90 ± 2.38	1.72 ± 0.02	0.41	7.79 ± 0.98
PS-8	23.63 ± 1.47	1.67 ± 0.00	0.49	11.65 ± 0.72
PS-12	26.20 ± 5.52	1.60 ± 0.00	0.62	16.14 ± 3.40
PS-30	94.46 ± 16.21	1.52 ± 0.24	0.74	69.75 ± 11.97
VAPOR	16.39(58.65) ¹	1.50	0.78	12.70(45.75) ¹

^aModified G calculated from the initial ramp of CI values to the depth of the observed ice lens (approximately 15 mm) in vapor deposited sample, seen in Figure 4.

Table A.4: Summary of excavation efficiency metrics derived from cone penetration tests.							
Identifier	Work Done (J)	Displaced Volume (cm ³)	Specific Energy of Displacement (J/cm ³)	Excavated Mass m_{exc} (g)	Specific Energy of Excavation (J/cm ³)	Mass of Water m_{water} (g)	Excavation Efficiency η_{exc} (J/g)
Dry	0.01 ± 0.0	0.72 ± 0.01	0.02 ± 0.0	-	-	-	-
Gran-1	0.02 ± 0.0	0.68 ± 0.01	0.03 ± 0.01	-	-	-	-
Gran-2	0.03 ± 0.0	0.73 ± 0.01	0.04 ± 0.0	-	-	-	-
Gran-3	0.03 ± 0.0	0.70 ± 0.03	0.04 ± 0.0	-	-	-	-
Gran-4	0.03 ± 0.01	0.71 ± 0.02	0.04 ± 0.01	-	-	-	-
Gran-5	0.02 ± 0.0	0.72 ± 0.0	0.03 ± 0.0	-	-	-	-
Gran-8	0.03 ± 0.0	0.72 ± 0.05	0.05 ± 0.0	-	-	-	-
Gran-12	0.03 ± 0.01	0.72 ± 0.02	0.05 ± 0.01	-	-	-	-
Gran-30	0.14 ± 0.06	0.73 ± 0.01	0.20 ± 0.08	-	-	-	-
Cem-1	0.65 ± 0.06	0.72 ± 0.02	0.91 ± 0.11	26.33 ± 8.62	0.041 ± 0.014	0.26 ± 0.09	2.7 ± 0.96
Cem-2	4.15 ± 0.09	0.73 ± 0.0	5.67 ± 0.11	11.54 ± 3.66	0.57 ± 0.16	0.23 ± 0.07	19 ± 5.4
Cem-3	11.1 ± 1.2	0.71 ± 0.05	15.74 ± 1.35	5.57 ± 1.19	3.1 ± 0.65	0.17 ± 0.04	69 ± 15
Cem-5	2.54 ± 0.65	0.10 ± 0.02	25.72 ± 1.37	0.27 ± 0.08	14 ± 1.1	0.01 ± 0.00	190 ± 15
Cem-12	1.71 ± 0.27	0.04 ± 0.01	40.42 ± 5.04	0.10 ± 0.00	26 ± 4.1	0.01 ± 0.00	140 ± 23
PS-1	0.17 ± 0.03	0.73 ± 0.01	0.23 ± 0.04	50.88 ± 14.72	0.0063 ± 0.0026	0.53 ± 0.16	0.35 ± 0.16
PS-2	0.17 ± 0.01	0.74 ± 0.01	0.23 ± 0.01	40.35 ± 7.00	0.0073 ± 0.0012	0.81 ± 0.15	0.21 ± 0.037
PS-3	0.17 ± 0.02	0.72 ± 0.01	0.23 ± 0.03	72.58 ± 16.67	0.004 ± 0.0007	2.20 ± 0.49	0.077 ± 0.012
PS-4	0.16 ± 0.02	0.73 ± 0.02	0.22 ± 0.02	36.27 ± 20.99	0.0084 ± 0.0044	1.45 ± 0.85	0.12 ± 0.066
PS-5	0.20 ± 0.03	0.73 ± 0.01	0.27 ± 0.04	50.12 ± 11.02	0.0069 ± 0.0013	2.49 ± 0.54	0.081 ± 0.016
PS-8	0.23 ± 0.02	0.73 ± 0.01	0.31 ± 0.03	38.18 ± 13.85	0.011 ± 0.0027	3.05 ± 1.11	0.079 ± 0.02
PS-12	0.22 ± 0.04	0.74 ± 0.01	0.29 ± 0.06	30.49 ± 8.16	0.012 ± 0.0041	3.66 ± 0.98	0.063 ± 0.022
PS-30	0.80 ± 0.22	0.75 ± 0.00	1.07 ± 0.29	22.55 ± 4.27	0.05 ± 0.015	6.76 ± 1.28	0.11 ± 0.052
Vapor	0.32 ± 0.00	0.74 ± 0.00	0.43 ± 0.00	-	-	-	-

Note: Metrics marked with “-” indicate data not available due to lack of measured mass or other physical parameters.

A.3.5 Specific Energy Extra

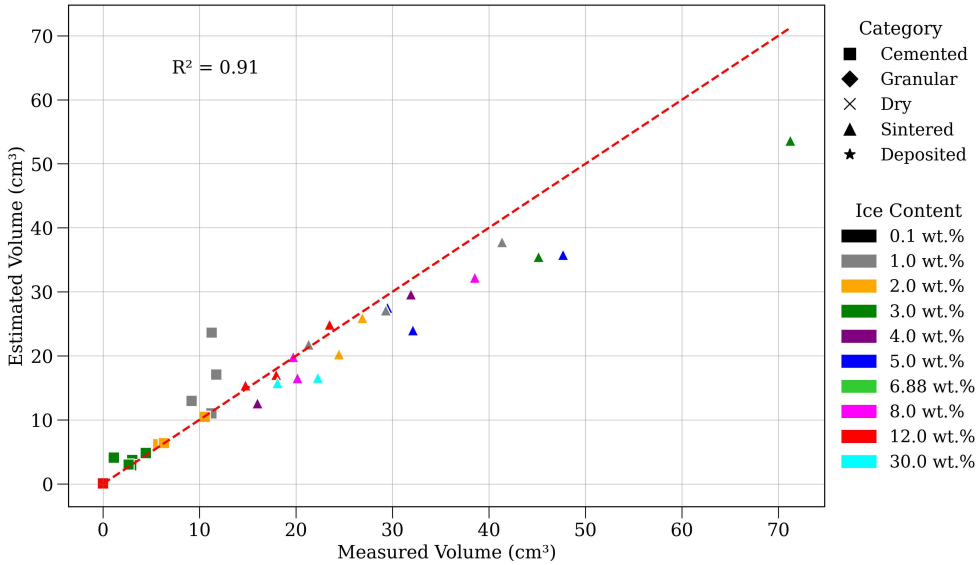


Fig. A7: Comparison of excavated volume estimated from LIDAR scans, or from measurements using known values of mass and density.

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